Foundation for Progress: Large Labeled Datasets and Hardware Acceleration

Danna Gurari
University of Colorado Boulder
Spring 2026

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https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last lecture:
  - The quest for deeper neural networks: why do they fail?
  - Convergence is hard: He initialization, batch normalization, Adam optimization
  - More is needed: residual learning with skip connections (ResNet)
  - Programming tutorial

- Assignments (Canvas)
  - Lab assignment 1 due in 1 week
    - For clarifications about common questions, review the Canvas assignment page

- Questions?
  - Beyond questions for me, please post them to Piazza and/or attend office hours

Today’s Topics

- The quest for more data and faster machines
- Paradigm shift: learned representations thrive with large labelled datasets
- Lucky break: gradient descent meets gaming hardware
- Tipping point: open-source platforms catalyze widespread engagement
- Programming tutorial

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Context: Early Acknowledgments of Data and Compute Bottlenecks

CNN backprop paper (LeNet-v1)

"Because training time was prohibitive (18 days on an Alliant mini-supercomputer)… our training times were "only" 3 days on a Sun workstation."

1989

2007

LeCun & Bengio paper

"The last quarter century has given us flexible non-parametric learning algorithms that can learn any continuous input-output mapping, *provided* enough computing resources and training data."

2025: Retrospective Thoughts



1947 – Present
Professor; Google Brain
2024 Nobel Prize winner

"In 1986, we showed that the backpropagation algorithm could learn the senses of words in a very simple toy example. We were very optimistic. We thought we had figured out how to make systems learn layers of features, how to make them learn to do vision, and that we would be able to make them learn to do language. We believed we had solved it and that everything would be wonderful…

That was in the 1980s and early 1990s. However, it would not scale up to recognizing real objects in real images, such as identifying a bird whether it was a seagull in the distance or an ostrich up close… We did not know at that time what the problem was. The problem was mainly that we did not have enough data or enough compute power… In fact, what we really needed was a million times as much data and a million times as much compute, and then it worked very well."

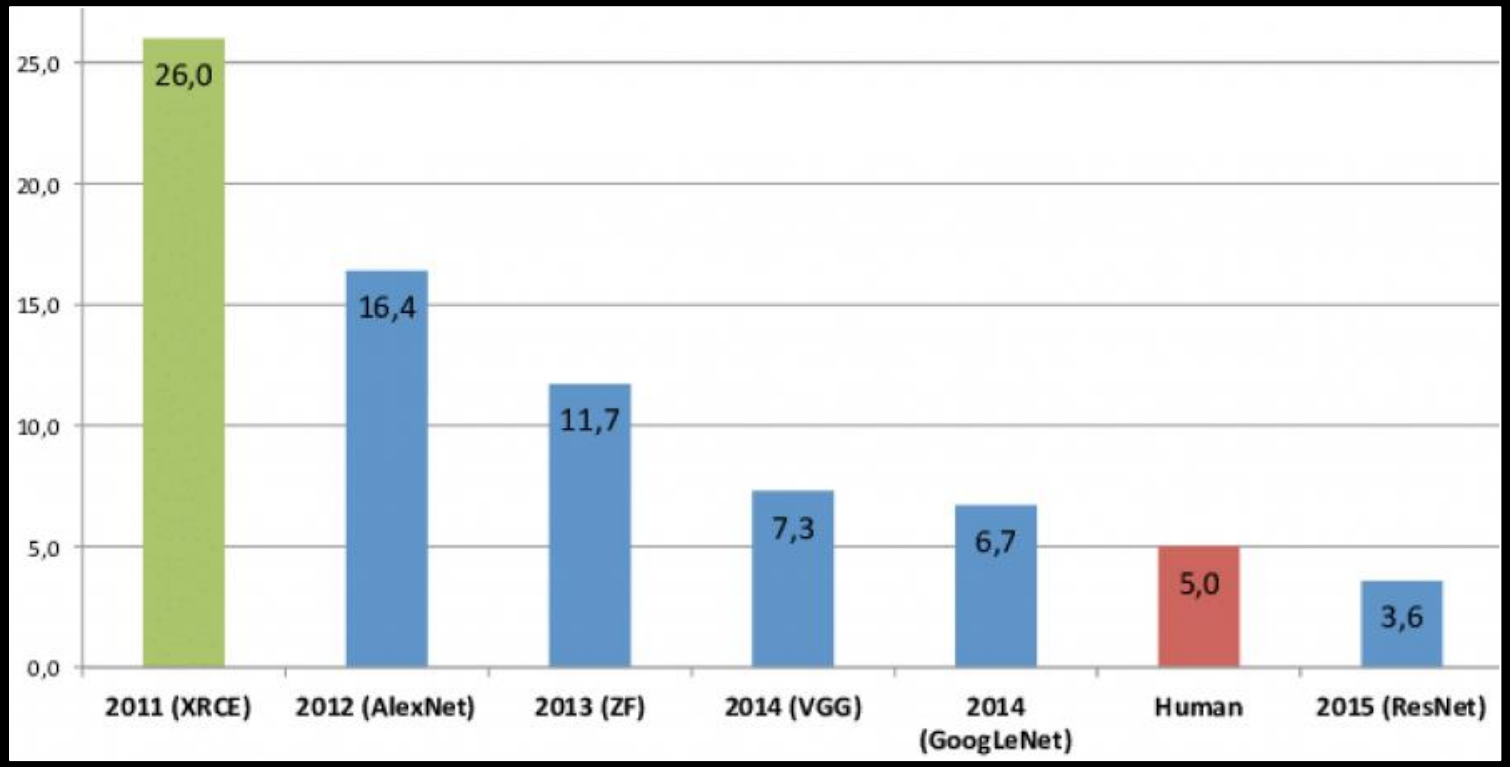
https://www.persuasion.community/p/geoffrey-hinton

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Recall: Learned Representations Overtook Feature Engineering in ImageNet Challenge

ImageNet, Top 5 Error



https://www.edge-ai-vision.com/2018/07/deep-learning-in-five-and-a-half-minutes/

Large amounts of ImageNet supervision was critical for neural networks to dominate

(CVPR, 2009)

ImageNet: A Large-Scale Hierarchical Image Database

Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li and Li Fei-Fei
Dept. of Computer Science, Princeton University, USA

ImageNet Context

Model Dataset



Model Dataset



Model Dataset

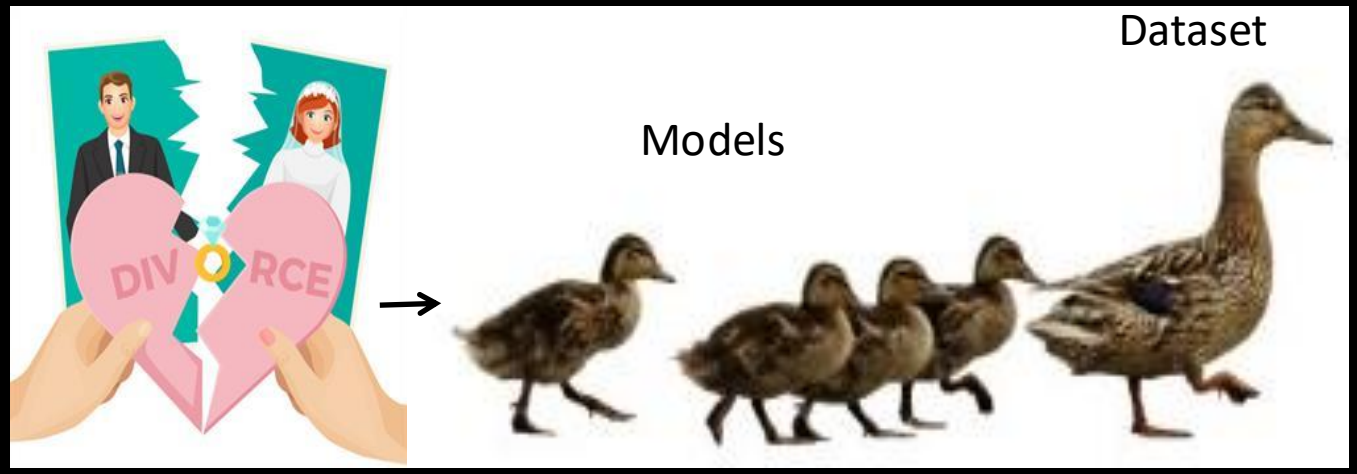


Model Dataset



Wave 1 of AI: Author Generated

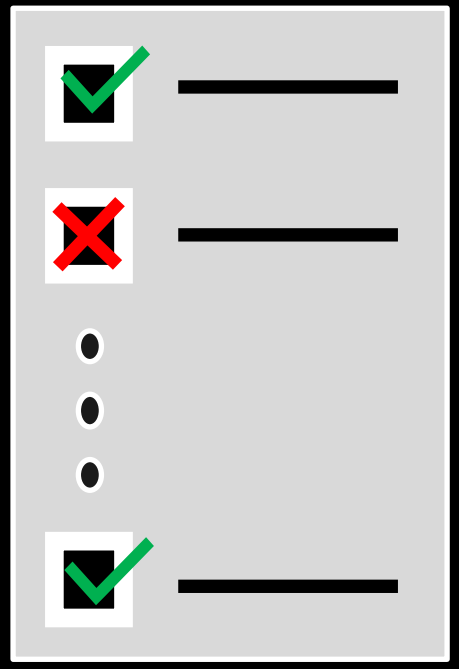
(Very Small Scale Due to Very Limited HW)



Waves 2 and 3 of AI: Standardized Benchmarks

(Larger Scale with Greater Diversity)

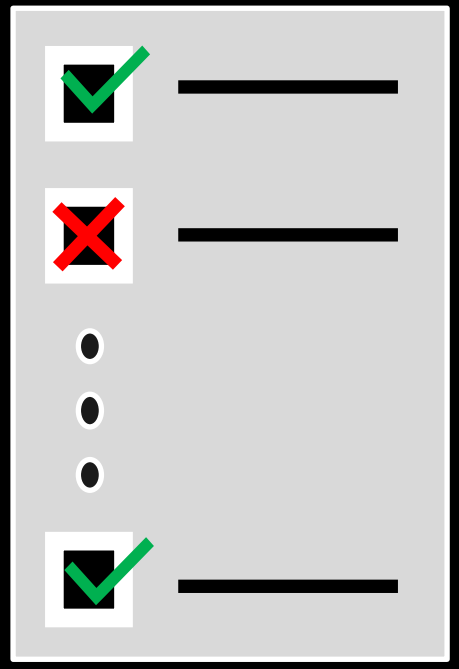
ImageNet Context



Essential ingredients for each benchmark:

1. Test examples that includes target results
2. Metric for assessing the similarity between each model prediction and the target result
3. New challenges for the community to tackle, evidenced by dataset analysis and model benchmarking

ImageNet Context



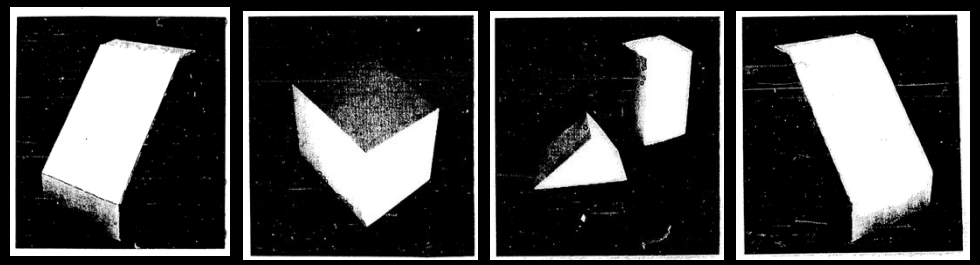


(Analogous to Tests with, Optionally, School Lessons)

ImageNet Context

1945 1963 1994 2009

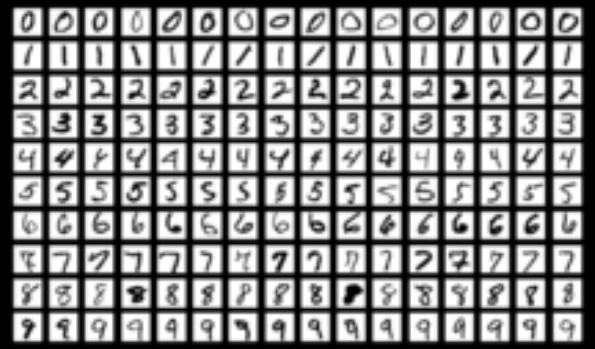




Roberts. Machine perception of three-dimensional solids. 1963

4 256x256 images could be stored to memory; scanning each to the computer took ~3 minutes

MNIST



70,000 28x28 images

ImageNet



3,200,000 images of varying resolution (e.g., 640 x 480)

Deng et al. 2009

ImageNet: Much Bigger with More Diversity

Hear from pioneer, Fei-Fei Li, about her gamble as an early career Assistant Professor to create ImageNet:



Video (5:44 – 10:10): https://youtu.be/40riCqvRoMs?feature=shared&t=343

Paradigm Shift Following ImageNet: Mimickry



https://3dvision.princeton.edu/projects/2010/SUN/

Scene Type Classification (2010)



https://cocodataset.org/#home

Object Localization (2014)



https://visualqa.org/

Visual Question Answering (2015)

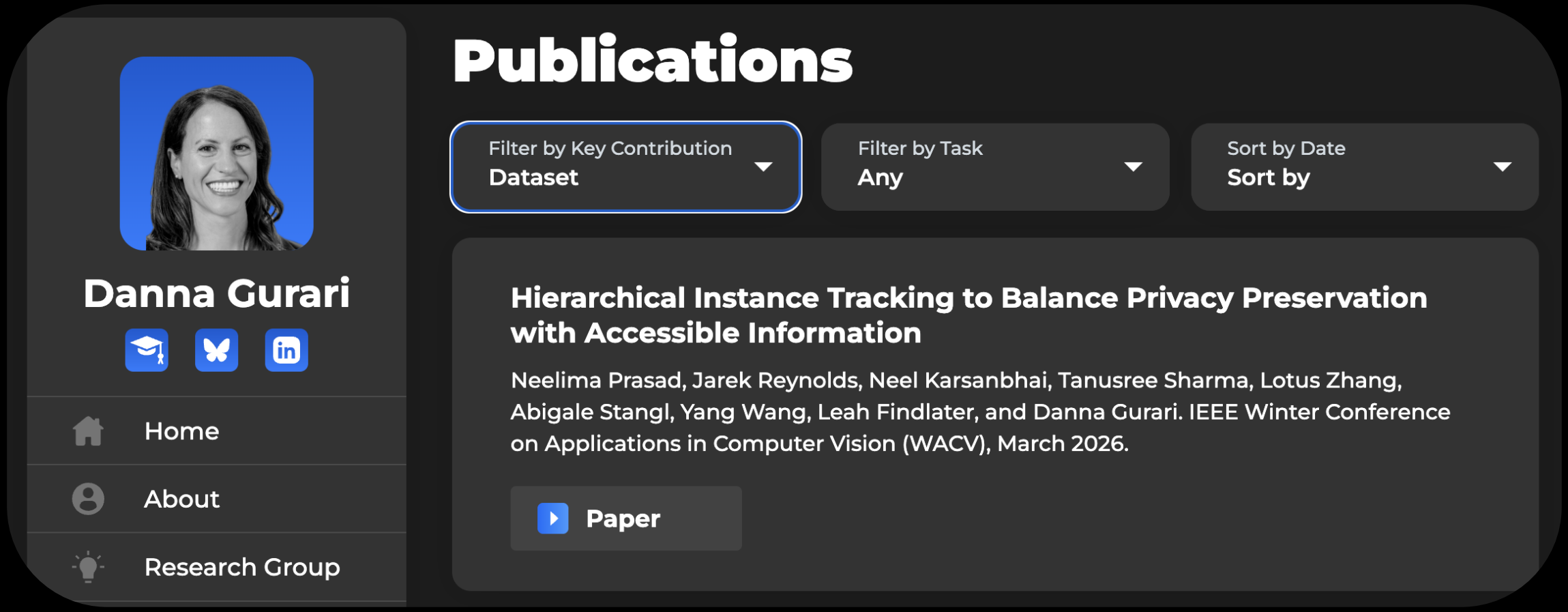


https://motchallenge.net/

Multi-Object Tracking (2015)

A surge of of new large labeled datasets emerged for many tasks,
followed by neural networks overtaking feature engineered SOTA for each

Paradigm Shift Following ImageNet: Mimickry



This includes from my research team, which has created over 20 public benchmarks

Key Ingredients Enabling Big Labeled Datasets

1. Increasing amounts of data posted to the Internet; e.g., 

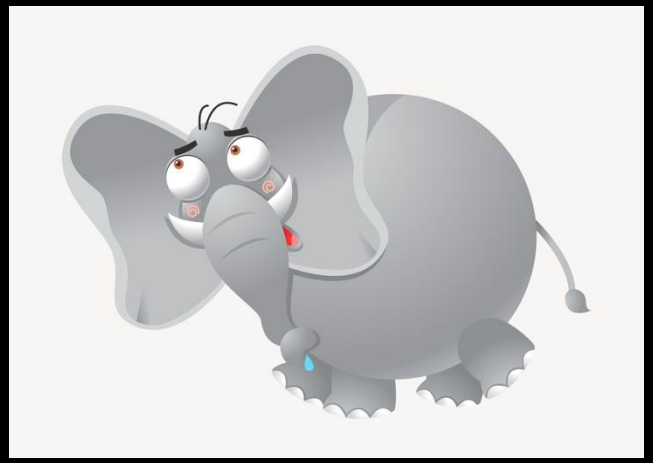
2. Greater hardware capacity; RAM (e.g., GBs) and storage (e.g., TBs)

3. Data annotation platforms providing on-demand workforce online

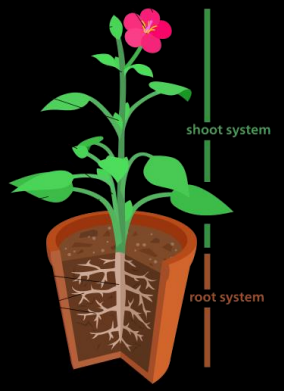
Data Annotation: Give it a Try!

- What object is in the image? (i.e., object recognition task)

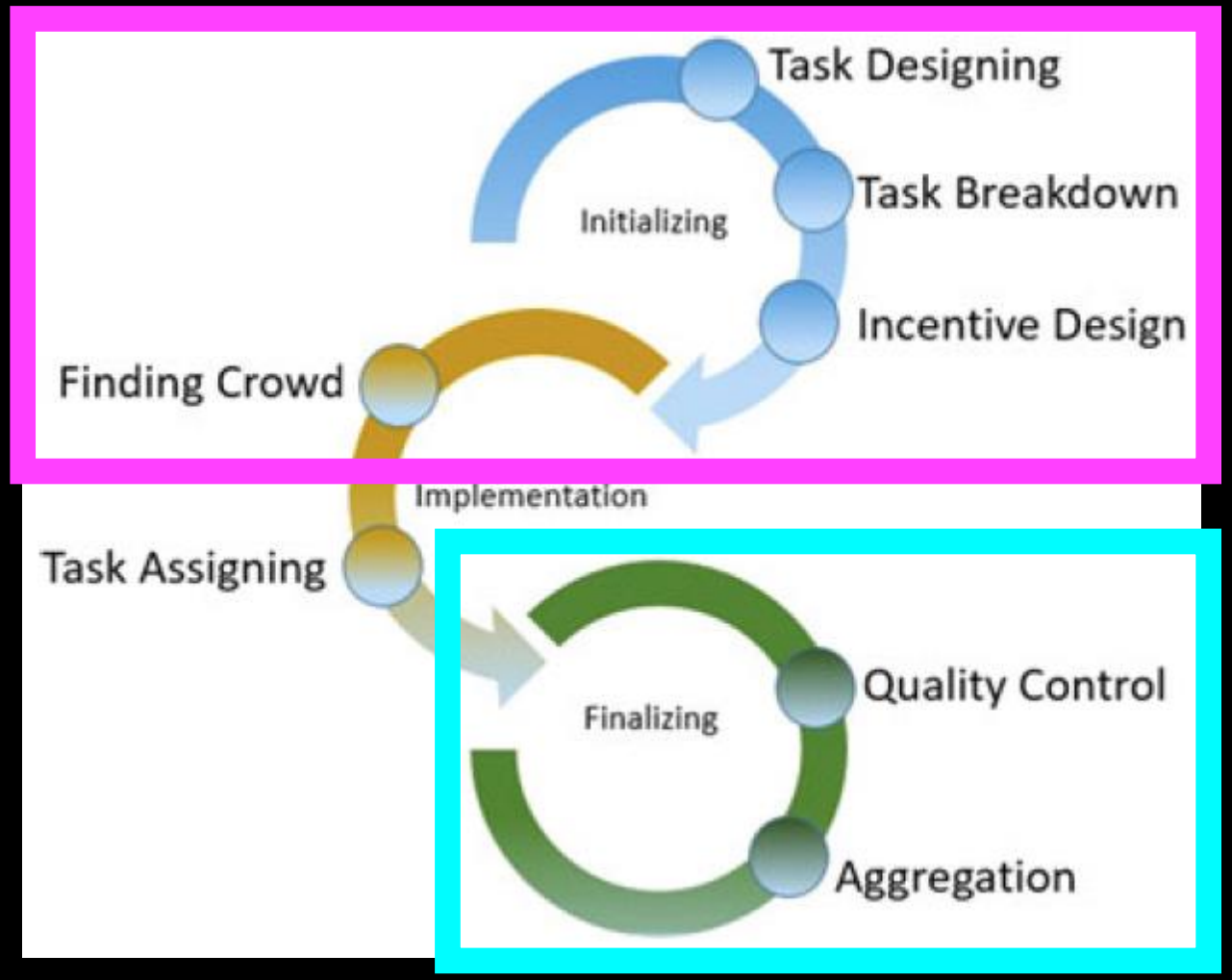








Data Annotation: How To Collect High Quality Annotations?



Pre-Task

- Instructions
- Incentives
- Required qualification criteria
- Training/Qualification tests

Post-Task

- Honey pot tasks
- Aggregate redundant responses
- Manual review
- Automated review

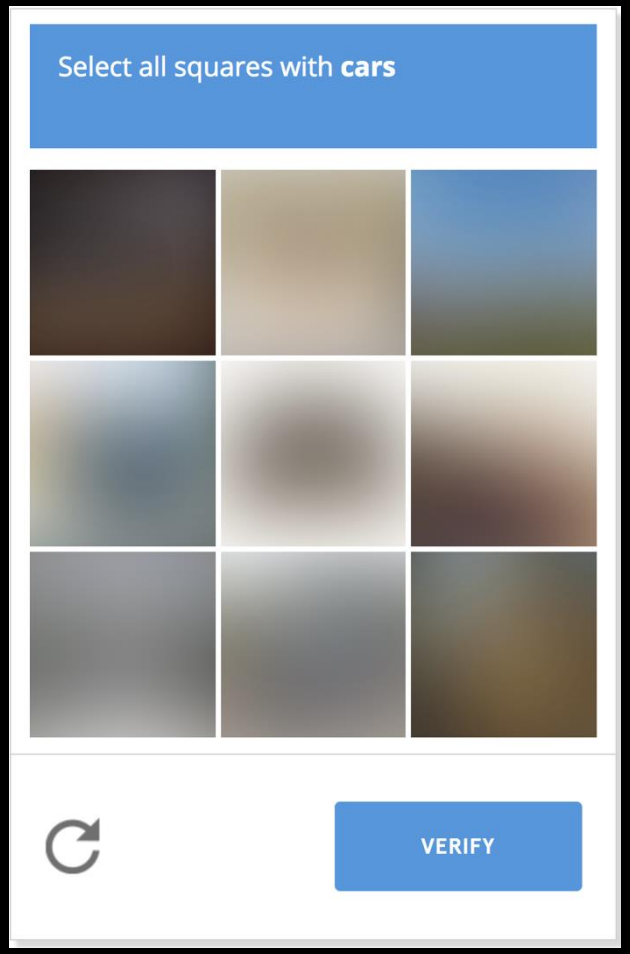
Chittilappilly et al. A Survey of General-Purpose Crowdsourcing Techniques. IEEE TKDE 2016.

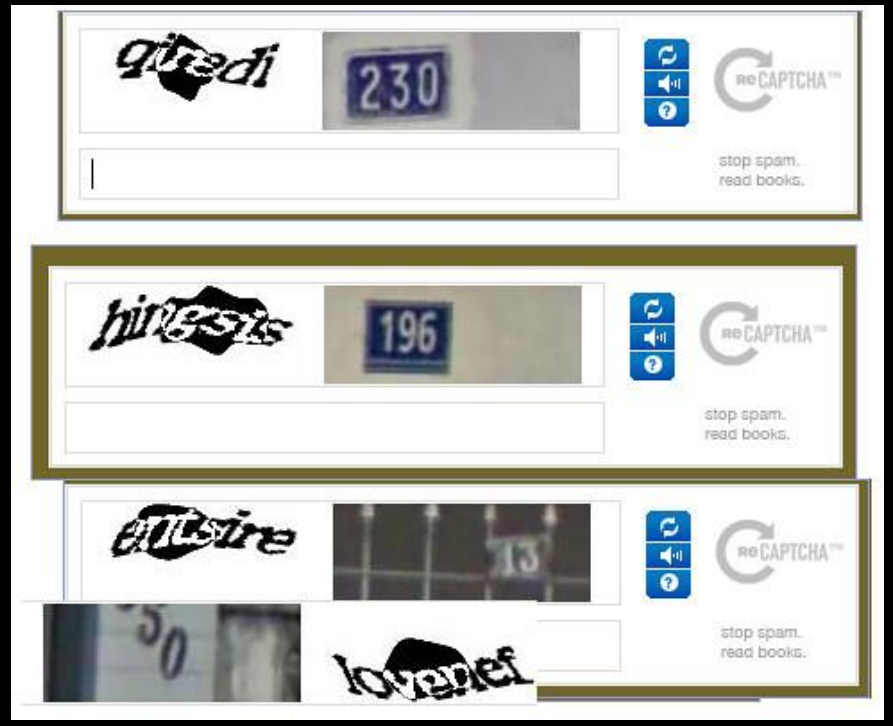
Data Annotation: How to Recruit Annotators?

- Security Questions
- Gamification
- Citizen Science
- Pay

Data Annotation: How to Recruit Annotators?

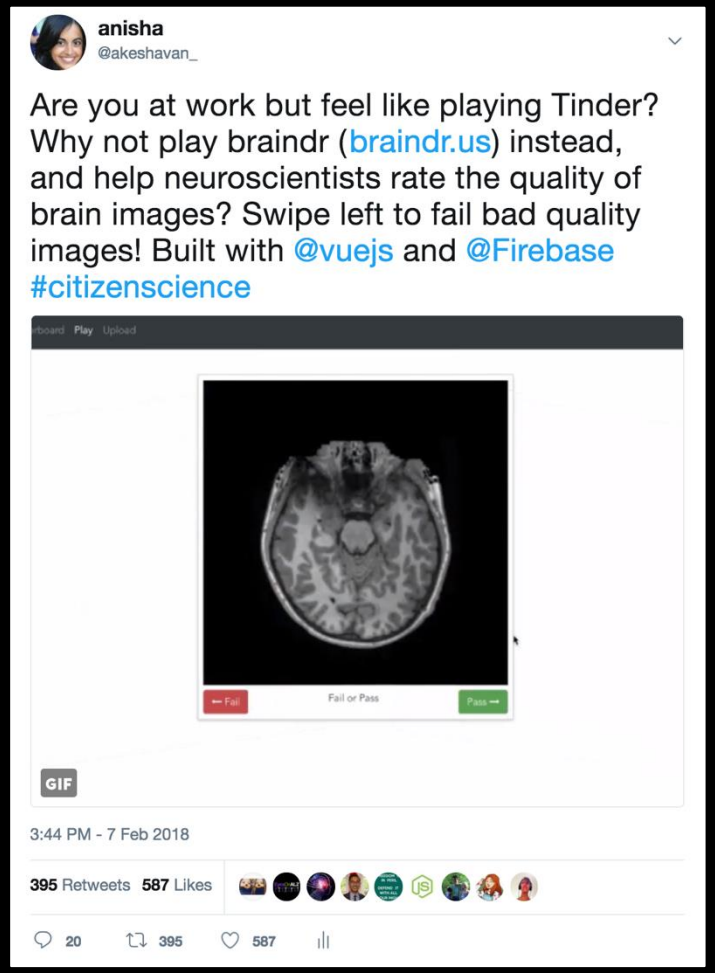
- Security Questions





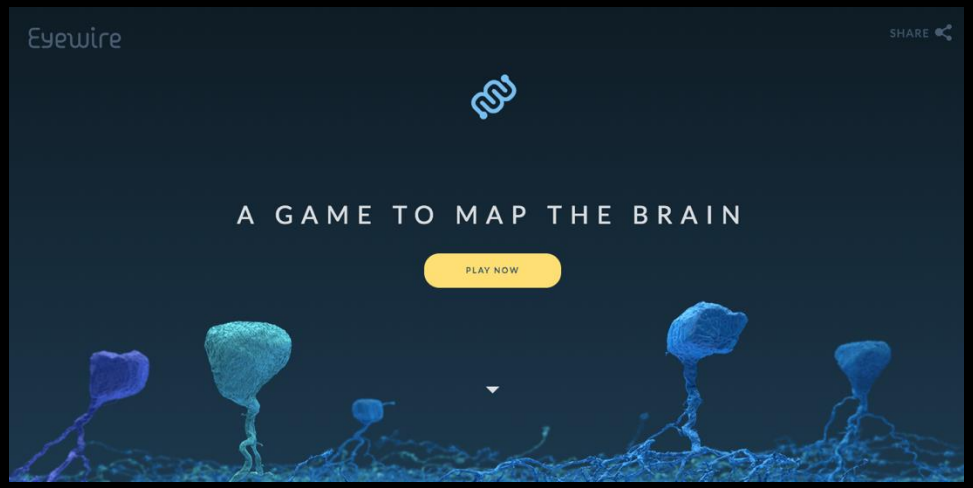
Data Annotation: How to Recruit Annotators?

- Gamification



[von Ahn & Dabbish; 2004]





Data Annotation: How to Recruit Annotators?

- Citizen Science



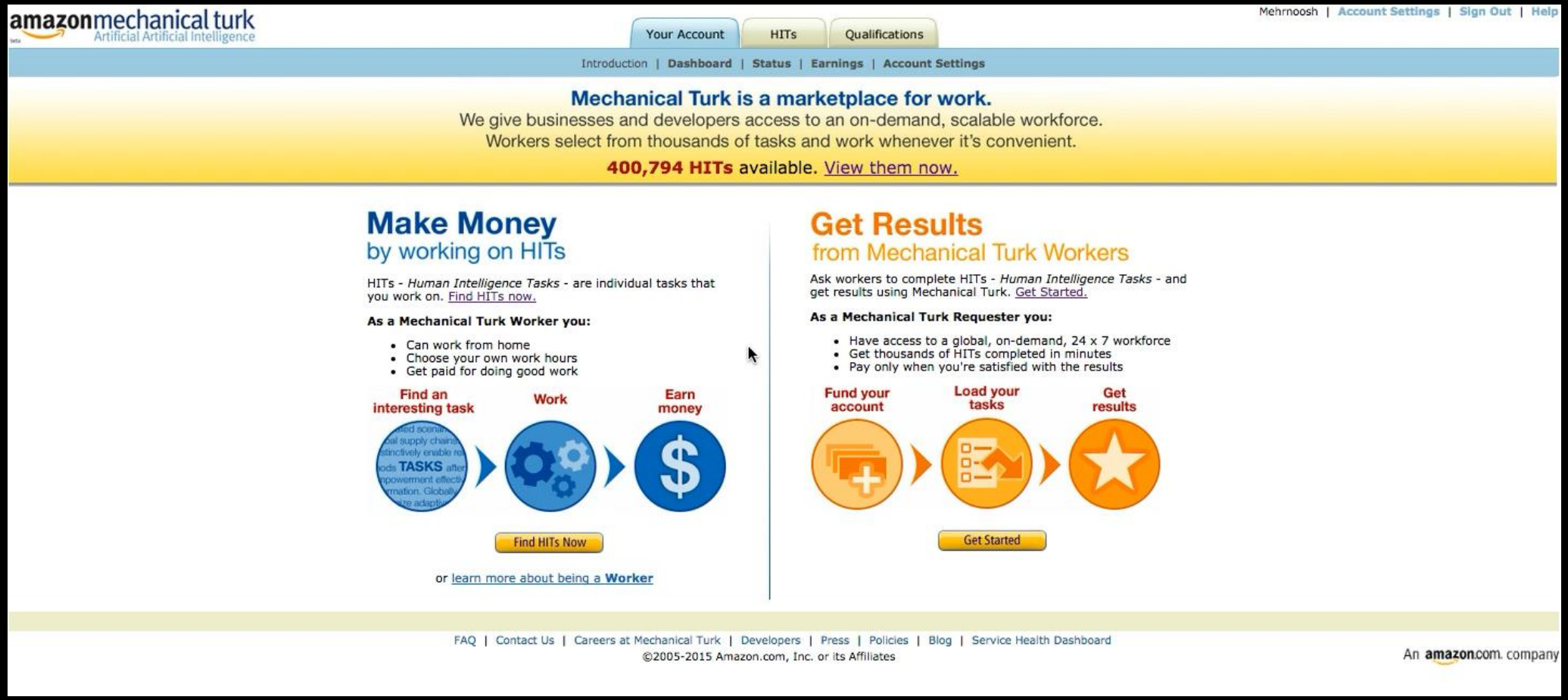
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Data Annotation: How to Recruit Annotators?

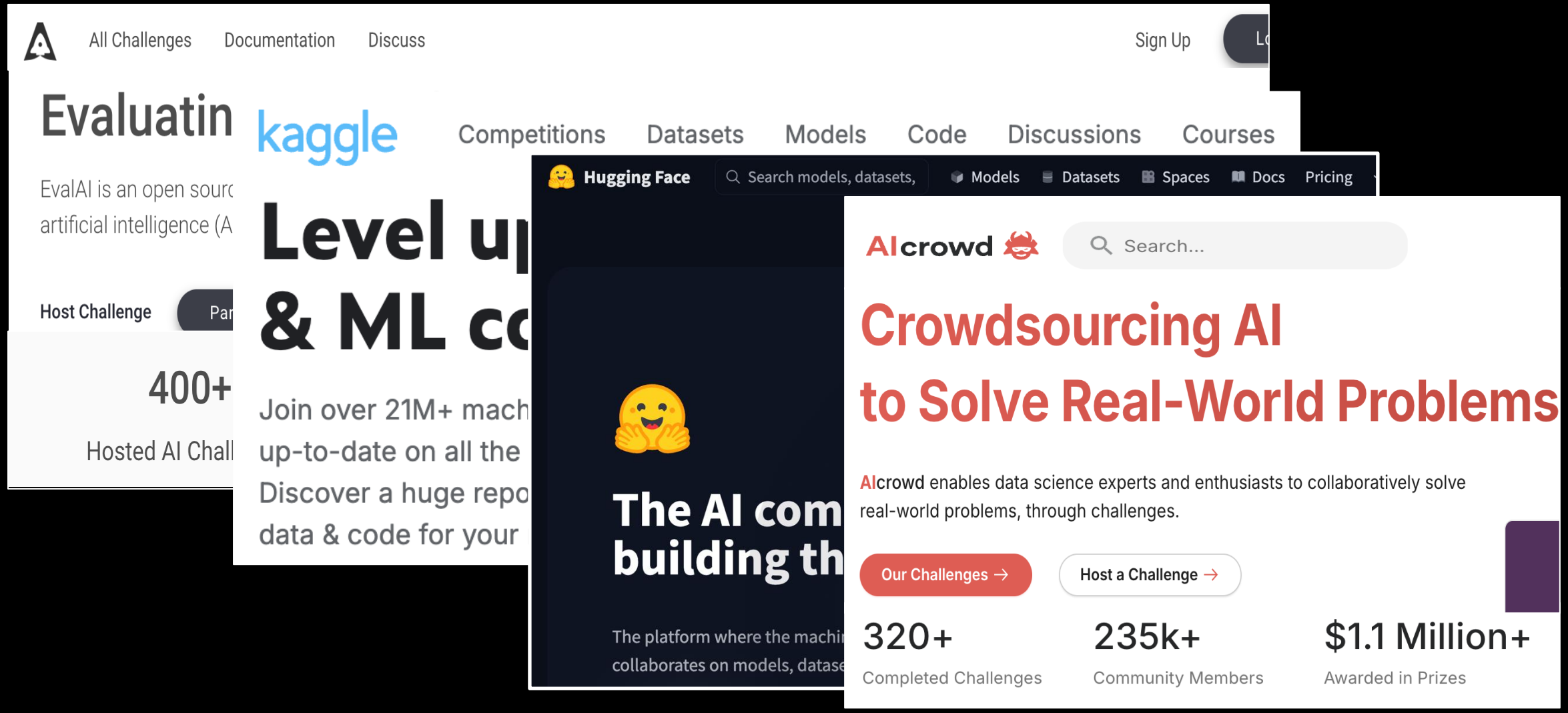
- Pay



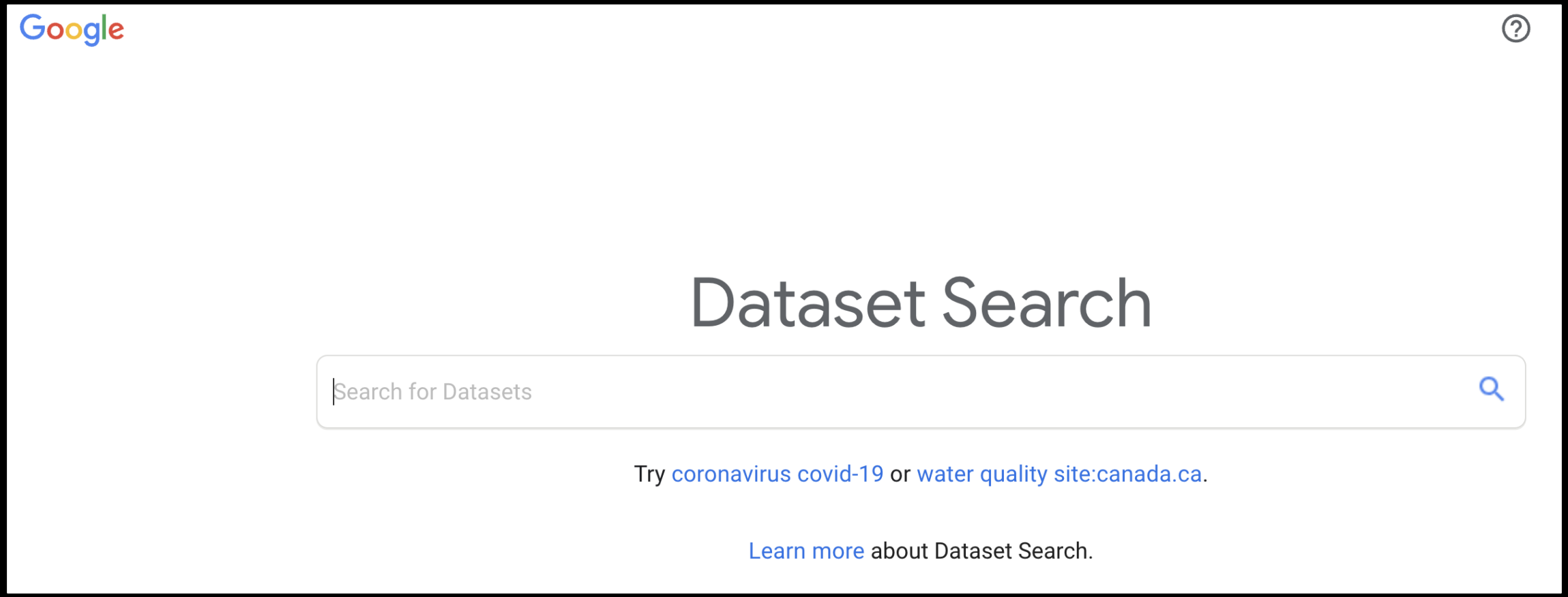
Which Annotators Would You Use When?

- Security Questions
- Gamification
- Citizen Science
- Pay

How to Find Dataset Challenges (Which Include Evaluation Servers)



How to Find Dataset Challenges (Without Evaluation Servers); e.g.,



ImageNet Impact Celebrated 10 Years Later



https://syncedreview.com/2019/06/18/cvpr-2019-attracts-9k-attendees-best-papers-announced-imagenet-honoured-10-years-later/

PAMI Longuet-Higgins Prize

Retrospective Most Impactful Paper from CVPR 2009

ImageNet: A large-scale hierarchical image database

Jia Deng, Wei Dong, Richard Socher,
Li-Jia Li, Kai Li, and Li Fei-Fei

"In 2009, ImageNet was not the most mainstream work, but all of us who did this project believed that it would have a big impact, so we put in a lot of efforts. One of the revelations it gives me is that you don't have to do the most popular things, but do what you believe will have an impact."

- First author, Jia Deng

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ImageNet: Necessitated New Breed of Hardware

- ImageNet's (inconveniently) large size: ~130 GB

- Typical HW in 2010 meant deep learning needed to shift away from local machines
  - RAM: 2-4 GB (stored less than 5% of ImageNet)
    - Consequence: MUST stream to disk and so bottleneck shifts to disk I/O
  - Storage: 160 - 500 GB (26% to 81% consumed by ImageNet)
    - Consequence: inconvenient local storage normalized a shift to external hard drives and institutional servers
  - Compute efficiency: high-end research machines had dual quad-core CPUs; researchers mainly used CPUs; cloud computing was emerging with CPUs
    - Consequence: training ImageNet on CPUs was prohibitively slow

- Does training deep neural networks make sense with your computer?
  - Does your computer have less than 40GB of RAM?
  - Does your computer have less than 1 TB?

Key Insight:

(ICML, 2009)

Large-scale Deep Unsupervised Learning using Graphics Processors

Rajat Raina RAJATR@CS.STANFORD.EDU
Anand Madhavan MANAND@STANFORD.EDU
Andrew Y. Ng ANG@CS.STANFORD.EDU
Computer Science Department, Stanford University, Stanford CA 94305 USA

"We argue that modern graphics processors far surpass the computational capabilities of multicore CPUs, and have the potential to revolutionize the applicability of deep unsupervised learning methods. We develop general principles for massively parallelizing unsupervised learning tasks using graphics processors… Our implementation of DBN learning is up to 70 times faster than a dual-core CPU implementation for large models. For example, we are able to reduce the time required to learn a four-layer DBN with 100 million free parameters from several weeks to around a single day."

Key Insight: Repurpose GPUs



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Central Processing Unit (CPU)

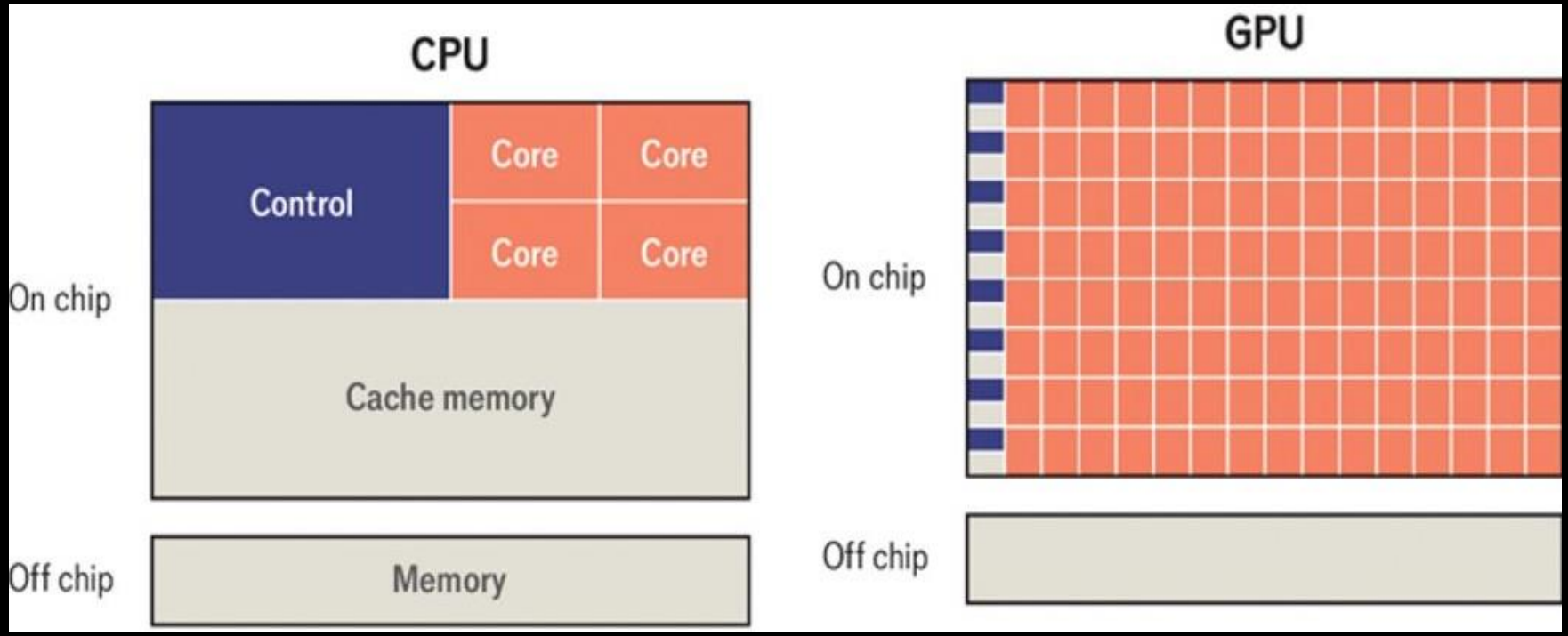


This image is in the public domain

Graphics Processing Units (GPUs)



Key Insight: Repurpose GPUs



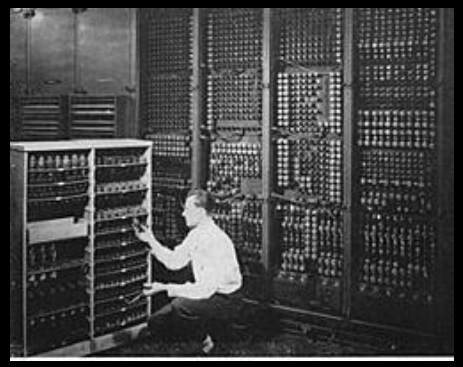
What are key architecture differences between CPUs and GPUS and their implications?

- Many more *parallel* processing cores accelerates workload
- Easier access to cache memory speeds up storing/retrieving intermediate computed results
- Simplified control focuses on scheduling parallel operations instead of managing *everything*

Source: Tsung Tai Yeh, National Yang Ming Chiao Tung University; The Economist ; September 21, 2024

Historical Context

1945 — First machine: room-sized since components were based on large vacuum tubes



(ENIAC's vacuum tubes being changed)

1959 — First silicon-based "chip": silicon components wired together on a single "integrated circuit" (IC)



1965 — Moore's law established: # of transistors on IC doubles every 2 (initially 1) years

The transistors switch on and off to generate the 0s and 1s of binary notation

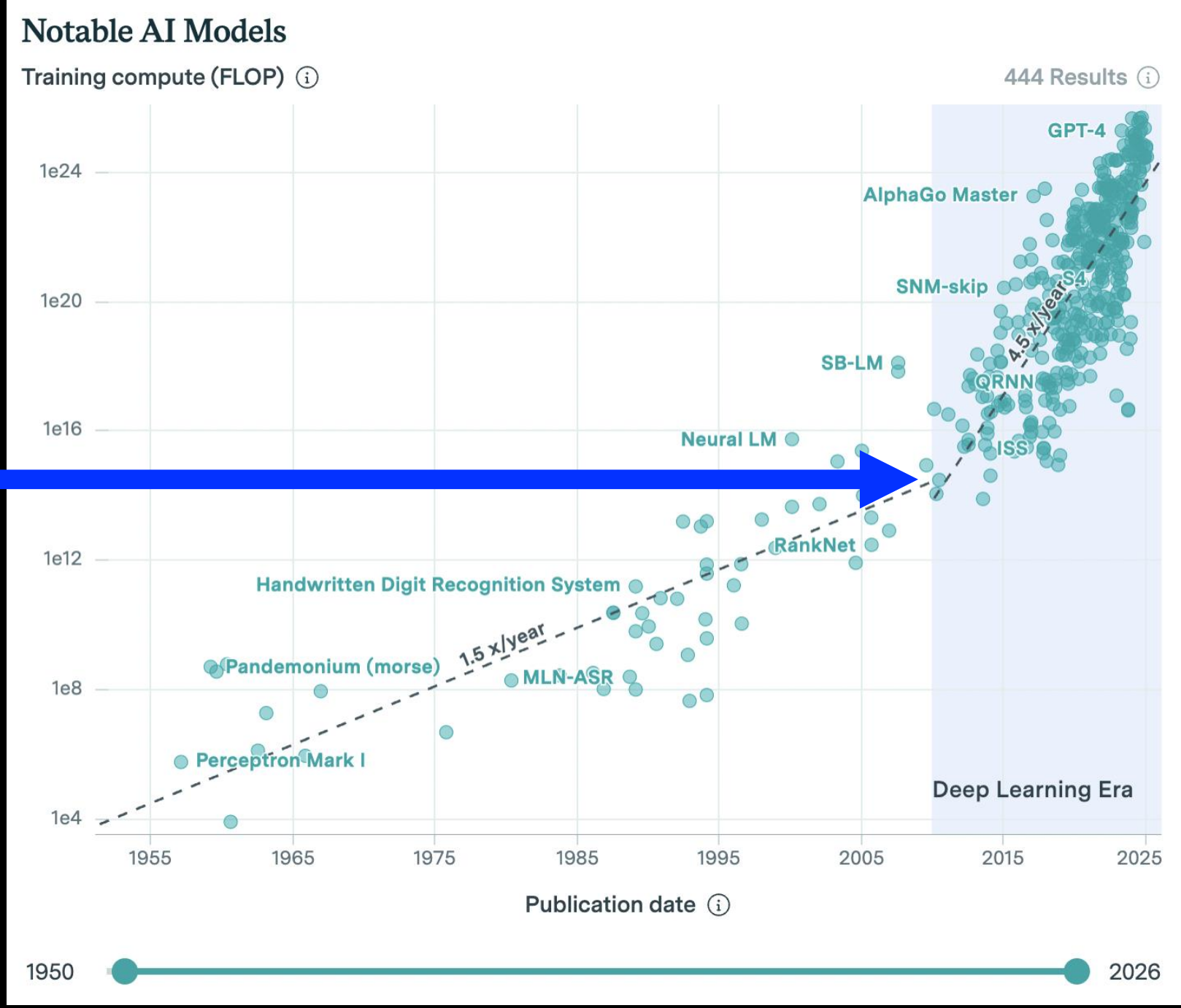
(e.g., 200 per mm^2 vs 150M per mm^2 in 1971 vs 2023 for Intel's MI300)

2010 — Computing for training AI models exceeded progress from Moore's law

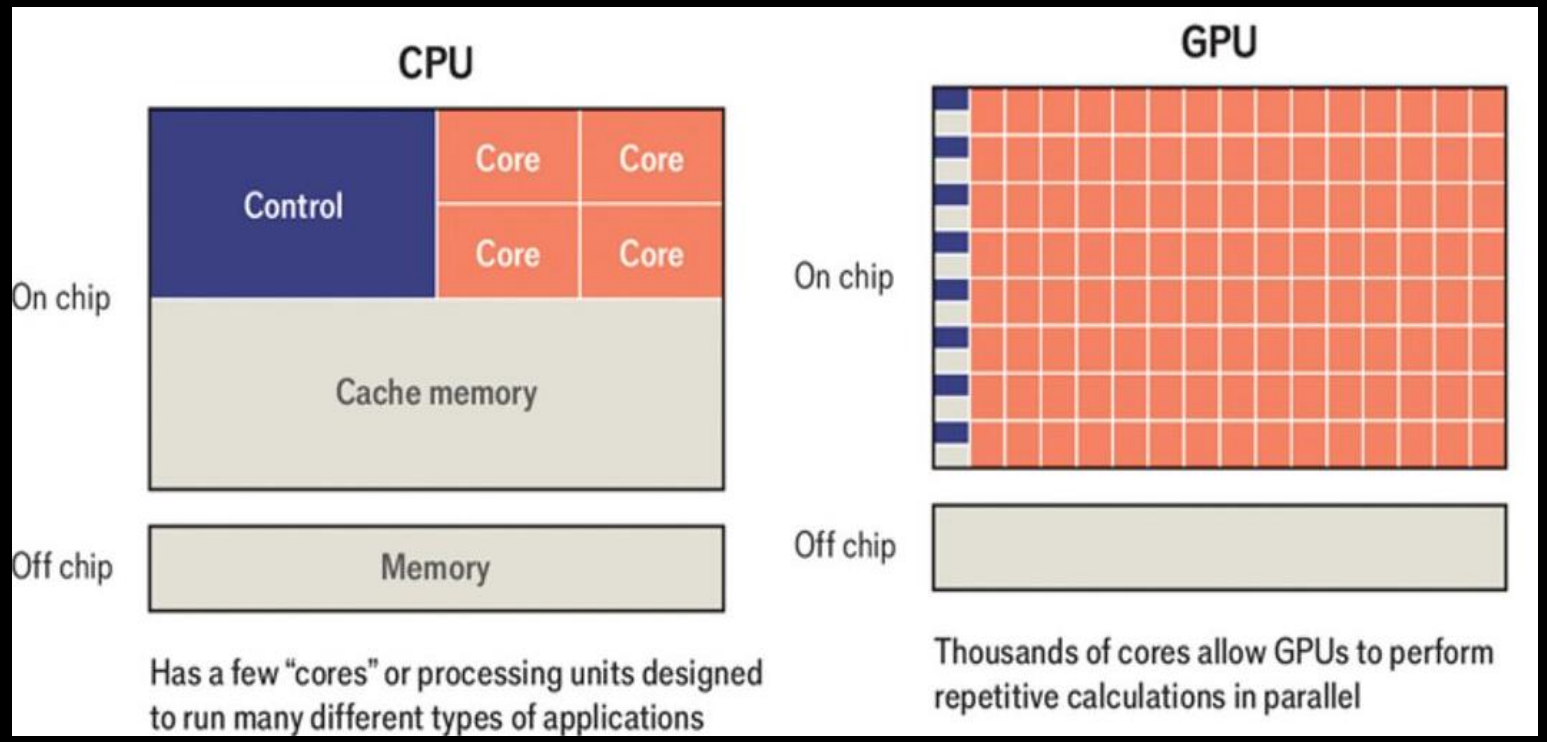
https://epoch.ai/blog/compute-trends; https://en.wikipedia.org/wiki/Invention_of_the_integrated_circuit

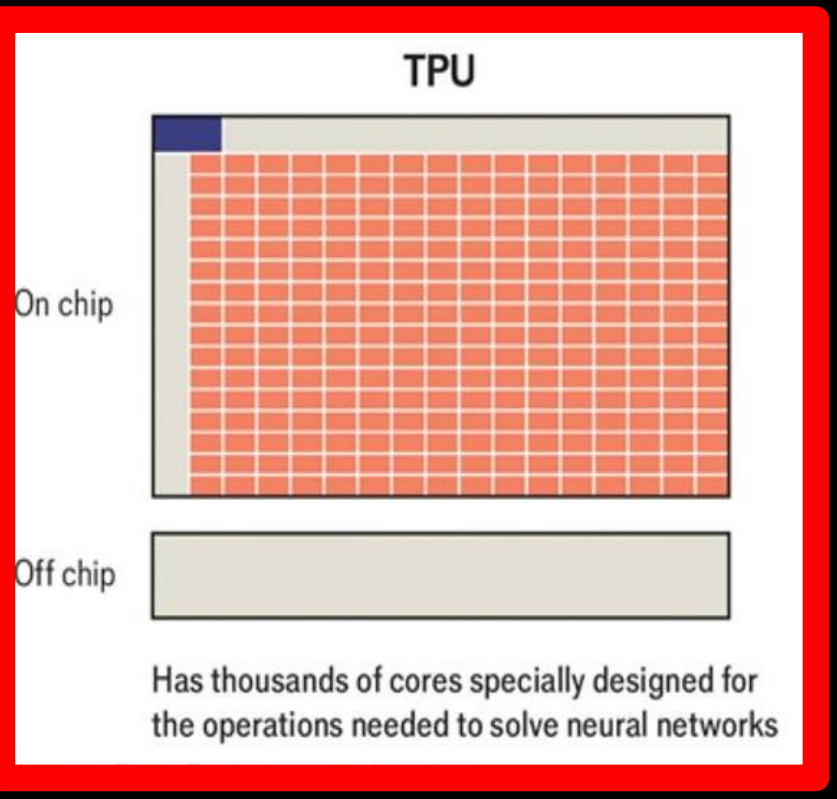
The Demand for Better HW

By 2012, the amount of computing needed to train AI models doubled every 6 months!



Rise of Specialized HW for Deep Learning; e.g.,



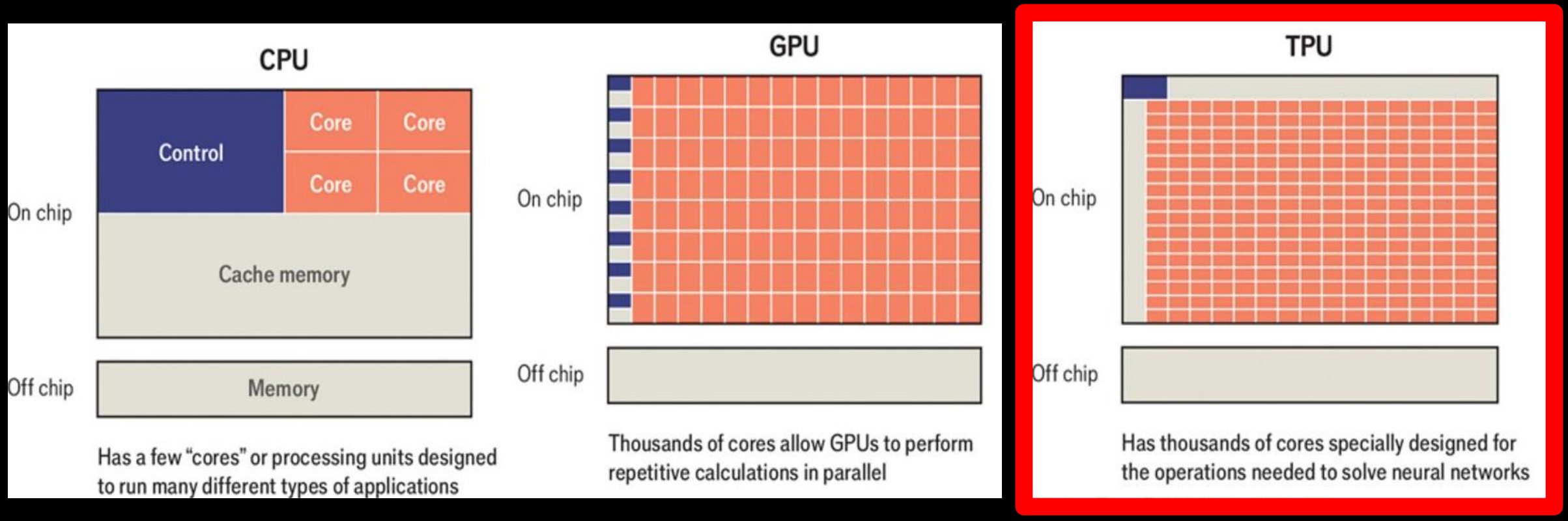


Introduced by Google for internal use in 2015 and public use in 2018

How do TPUs differ from GPUs and what are the implications?

Source: Tsung Tai Yeh, National Yang Ming Chiao Tung University; The Economist ; September 21, 2024

Rise of Specialized HW for Deep Learning; e.g.,



Flexibility

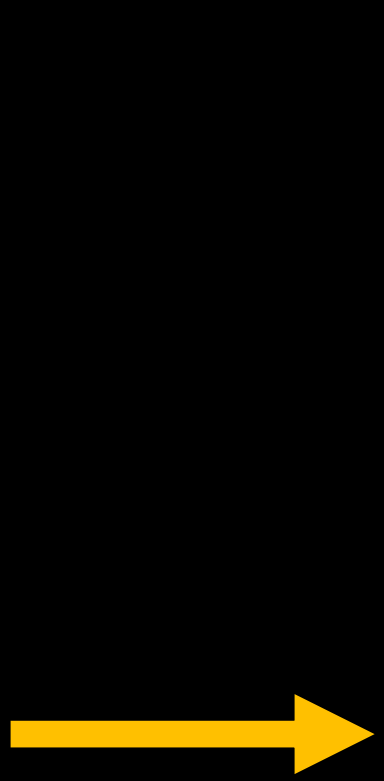
Efficiency (Faster & Less Energy Hungry)

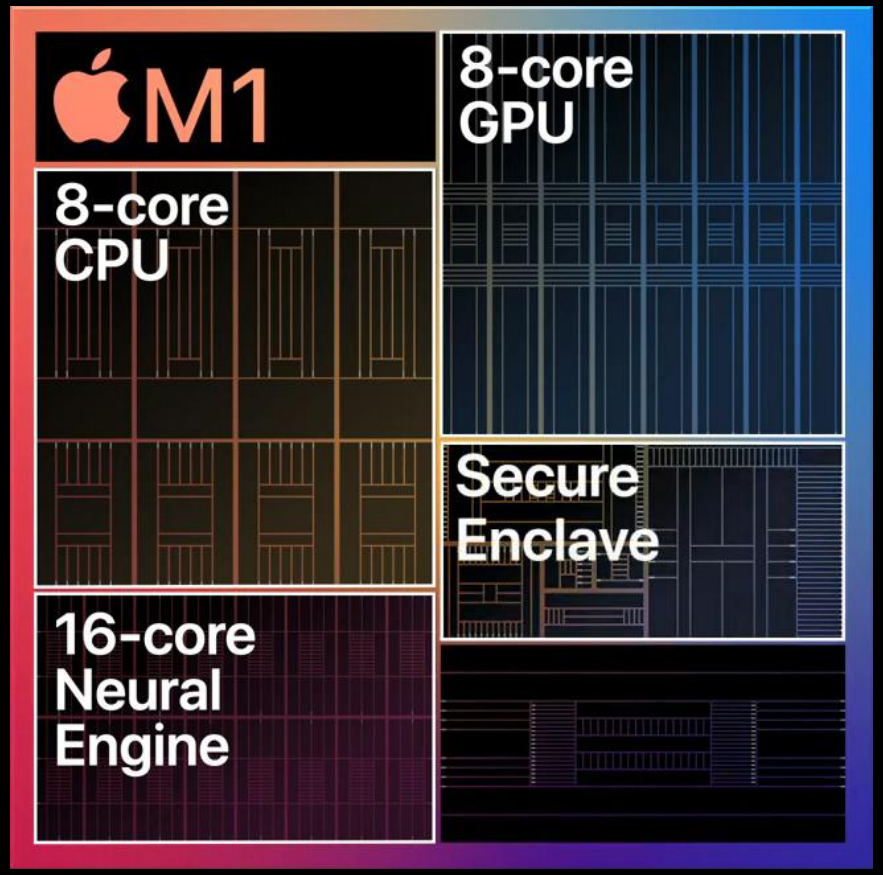
Source: Tsung Tai Yeh, National Yang Ming Chiao Tung University; The Economist ; September 21, 2024

Rise of Specialized HW for Deep Learning; e.g.,

- Alternatively, one chip can do it all (e.g., “Neural Processing Unit” or NPU)

Added by Apple in 2017 to their chips, a dedicated block for NN operations to benefit users of their devices (e.g., phones, iPads, laptops)

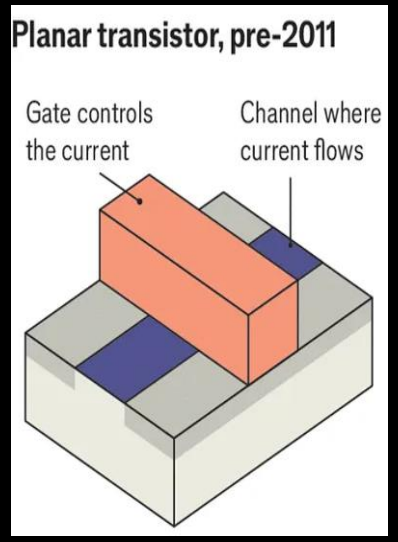




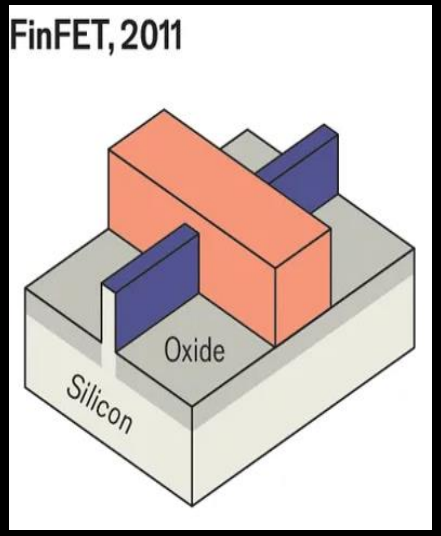
https://apple.fandom.com/wiki/Neural_Engine?file=Apple_M1_simplified_schematic.jpg

Rise of Specialized HW for Deep Learning

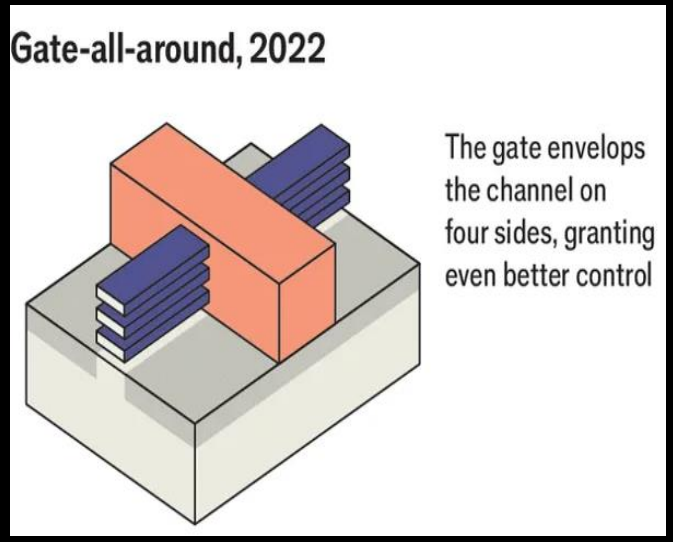
- Many companies are innovating to address the next key challenges:
  - Getting more transistors on a chip while minimizing energy consumption and waste heat
  - Less expensive factories to build chips (e.g., $20bn for one to be built in Arizona by TSMC)



Issue: current leaks in small transistors (e.g., 90 nm) from quantum effects, even when they are off



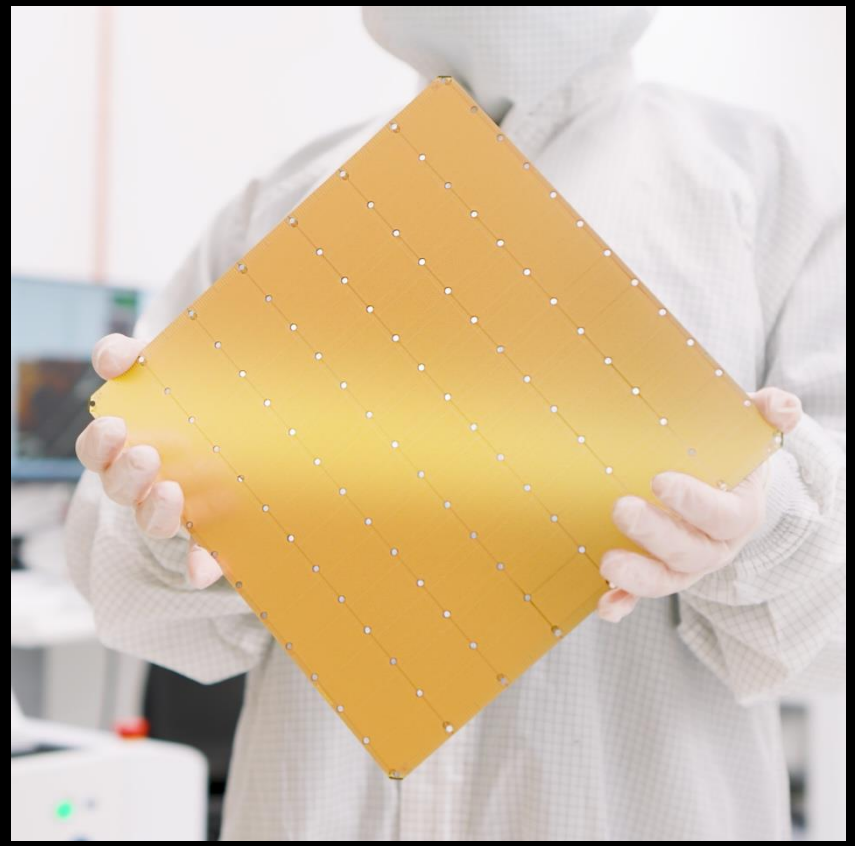
Introduced in 2011 by Intel, channels embedded in gates on 3 sides achieved better control of current flow and so less leakage



Introduced in 2022 by Samsung, gates surrounded channels on all sides to achieve better control of current flow

The Economist; September 21, 2024

State-of-the-Art for Single Chip (in the World!)

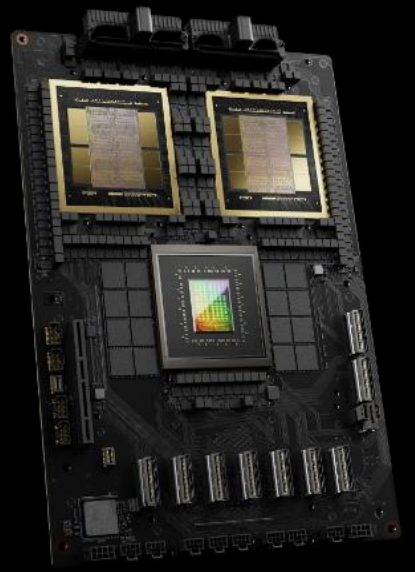


Cerebras WSE-3:

- Approximately size of a dinner plate, so all components can be on a single chip
- 900,000 cores
- 4 trillion transistors
- 44 GB of on-chip memory
- Rumored to cost $2-3 million

https://cerebras.ai/product-chip/

Blackwell: Nvidia’s Best Chip, Released in 2025



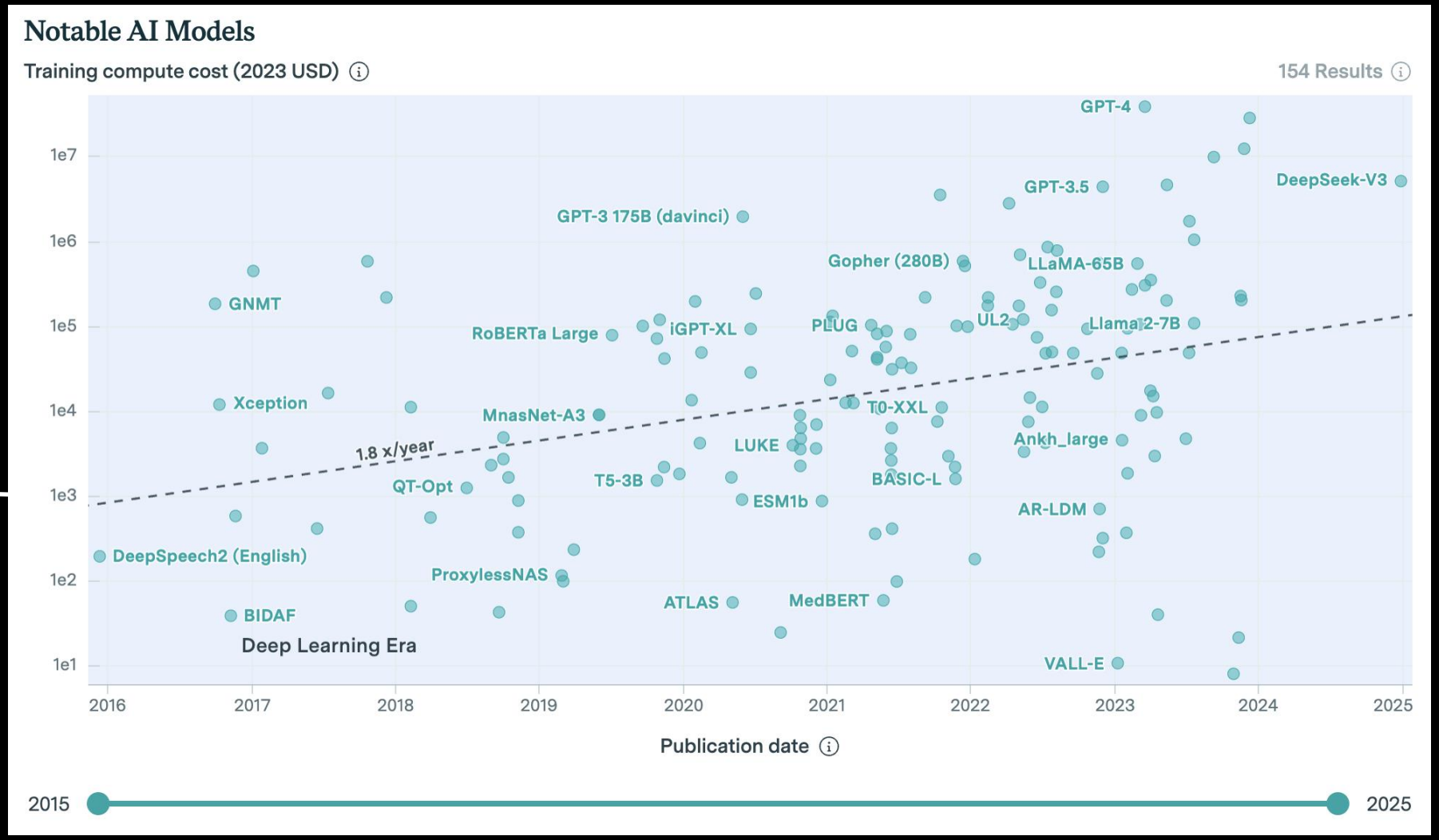
- 208 billion transistors on 2 connected dies
- Each die surrounded by 4 blocks of memory, totaling at 192 GB
- Over 1 year, it would consume ~1/2 the energy of an average American household (5.2 MWhrs)

Do you think a Blackwell costs (A) $4,000, (B) $40,000, or (C) $400,000

https://nvidianews.nvidia.com/news/nvidia-blackwell-platform-arrives-to-power-a-new-era-of-computing

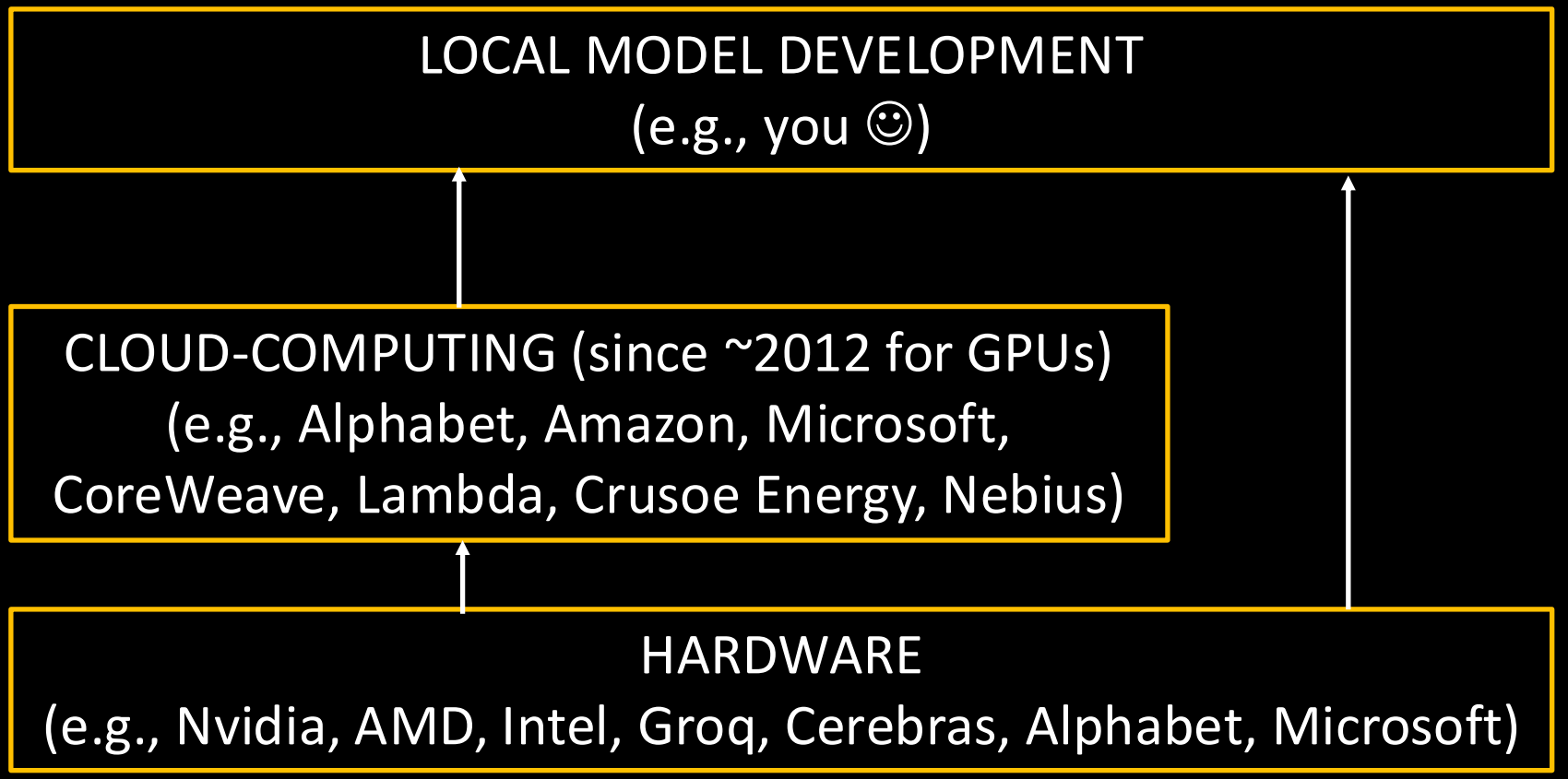
Compute Cost for Training Models from Scratch

e.g., $1000



Interactive Demo: https://epoch.ai/data/notable-ai-models#explore-the-data

How to Leverage GPUs? – Rent Versus Buy



e.g., in 2026, buy Nvidia's Blackwell at ~$40k versus rent at ~$20 per hour (~$13,440 for 4 weeks of 24/7 use)

Congratulations!

- By taking this class, you receive gifts of cloud credits worth over:



- Thanks to:

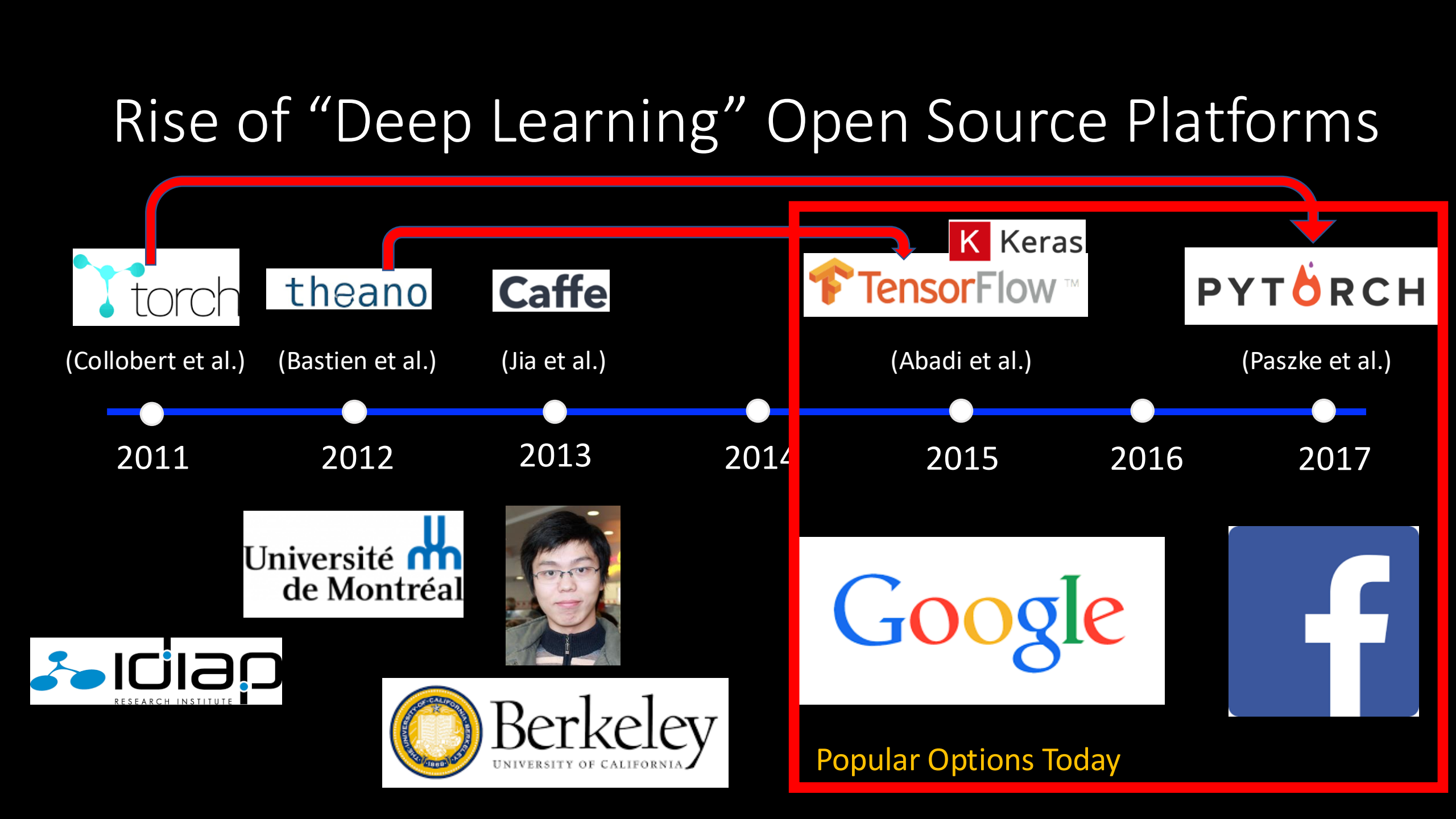
Jetstream2

Google

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- Paradigm shift: learned representations thrive with large labelled datasets
- Lucky break: gradient descent meets gaming hardware
- Tipping point: open-source platforms catalyze widespread engagement
- Programming tutorial

Rise of “Deep Learning” Open Source Platforms



Rise of “Deep Learning” Open Source Platforms

Course focus for programming tutorials

torch (Collobert et al.) — 2011

theano (Bastien et al.) — 2012

Caffe (Jia et al.) — 2013

Keras, TensorFlow™ (Abadi et al.) — 2015

PYTORCH (Paszke et al.) — 2017

2014 2016

Université de Montréal



idiap RESEARCH INSTITUTE

Berkeley UNIVERSITY OF CALIFORNIA

Google

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Platform Features Enabling Faster, More Efficient GPU Training

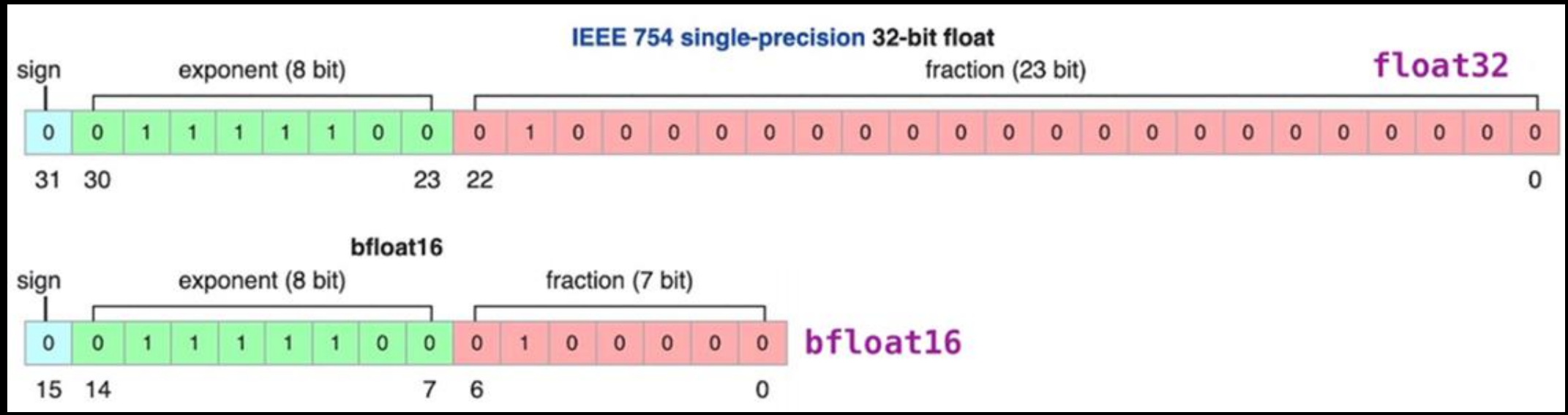
- Quantization: reduce precision
- Operator fusion: reduce memory read/writes
- Distributed optimization: parallelize computation

Platform Features Enabling Faster, More Efficient GPU Training

- Quantization: reduce precision
- Operator fusion: reduce memory read/writes
- Distributed optimization: parallelize computation

Quantization: Reduce Precision

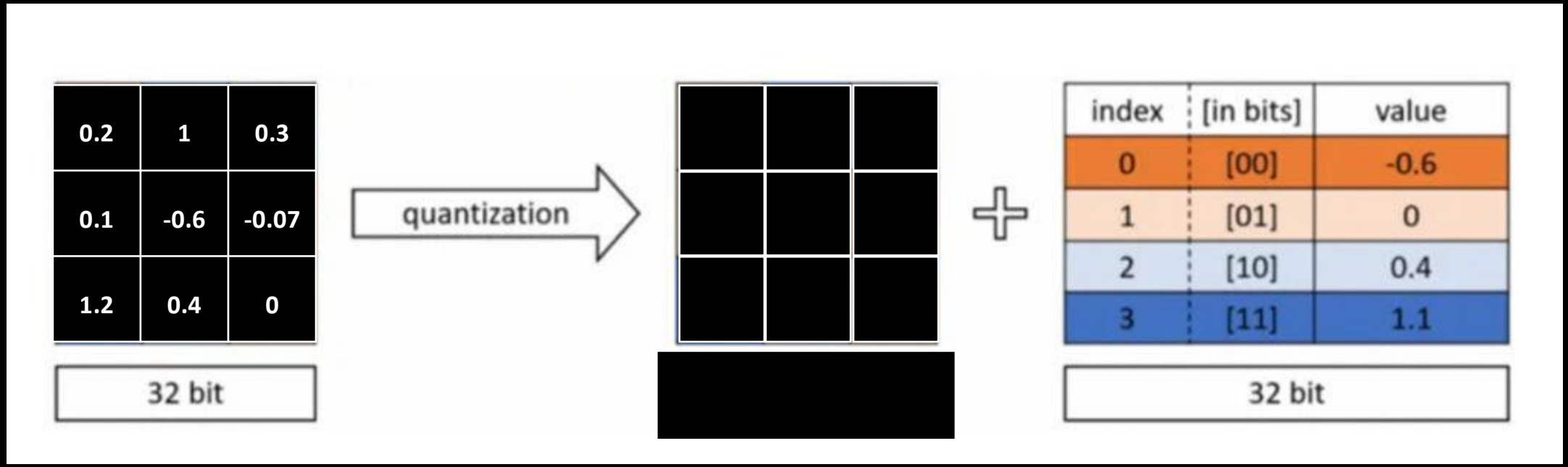
- Which precision and ranges do different data types offer?
  - 32 bits: e.g., ~ $\pm 3.4 \times 10^{38}$ with 2^{32} or ~4 billion values (default PyTorch type)
  - 8 bits: [-127, 127]
  - 4 bits: [-8, 7]



https://medium.com/@dillipprasad60/qlora-explained-a-deep-dive-into-parametric-efficient-fine-tuning-in-large-language-models-llms-c1a4794b1766

Quantization: Reduce Precision

- e.g.,
  - what number of bits are used in this example?
  - what bin value should be used for each of the 9 quantized values?

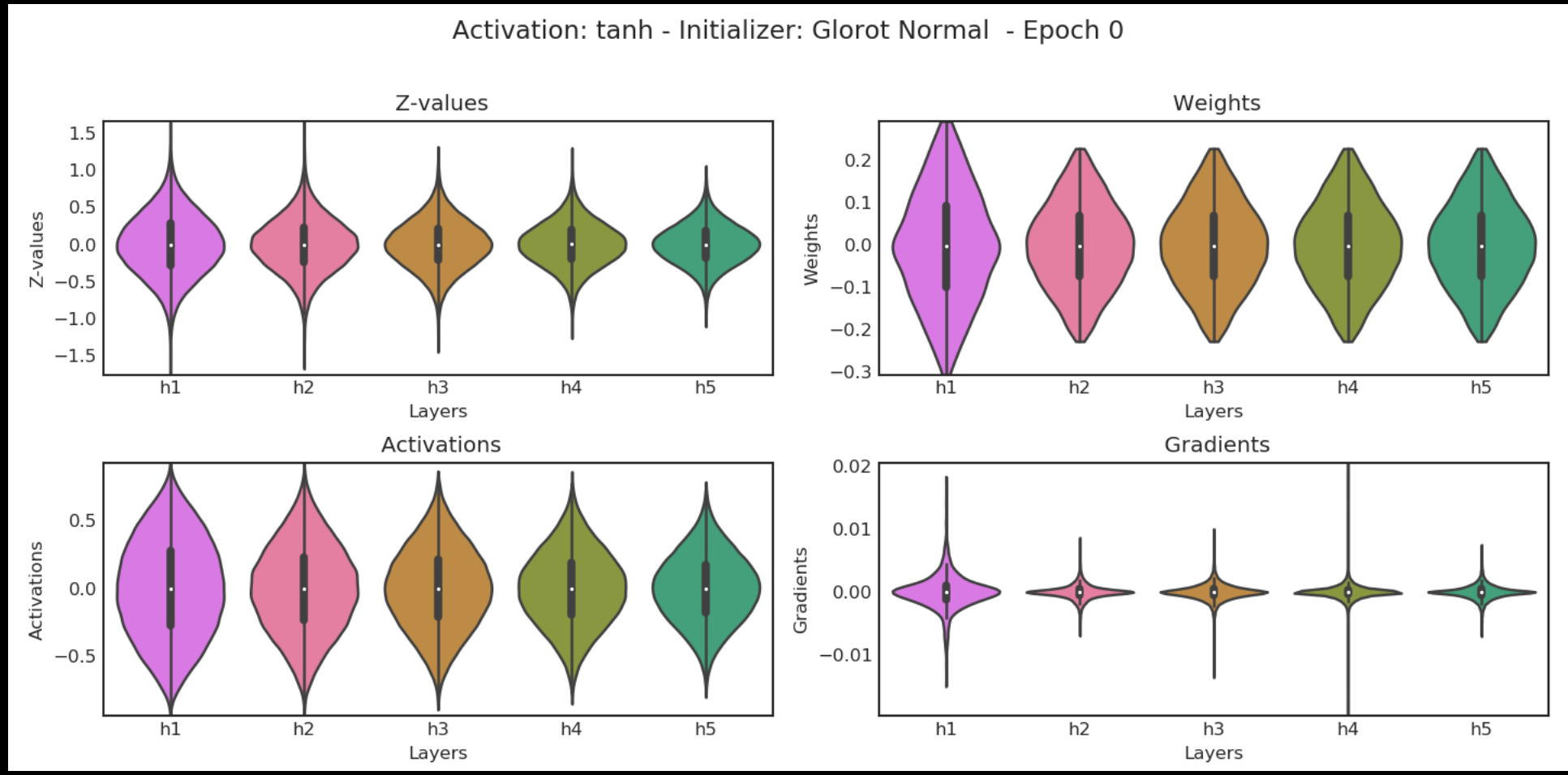


https://xailient.com/blog/4-popular-model-compression-techniques-explained/

Quantization at Training

How much precision is needed for training?

e.g., recall batch normalization causes values to center on 0 and range roughly between -1 and 1



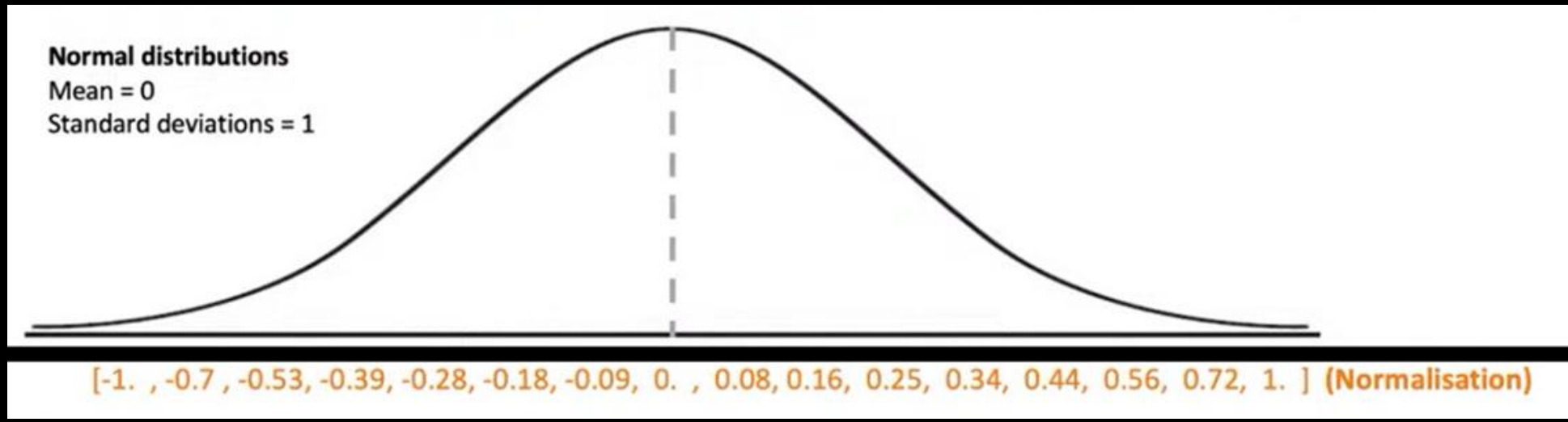
https://towardsdatascience.com/hyper-parameters-in-action-part-ii-weight-initializers-35aee1a28404

Quantization at Training

e.g., 4 bits means values mapped to 16 floating point bins (i.e., 2^4 possible values)

How much precision is needed for training?

e.g., recall batch normalization causes values to center on 0 and range roughly between -1 and 1



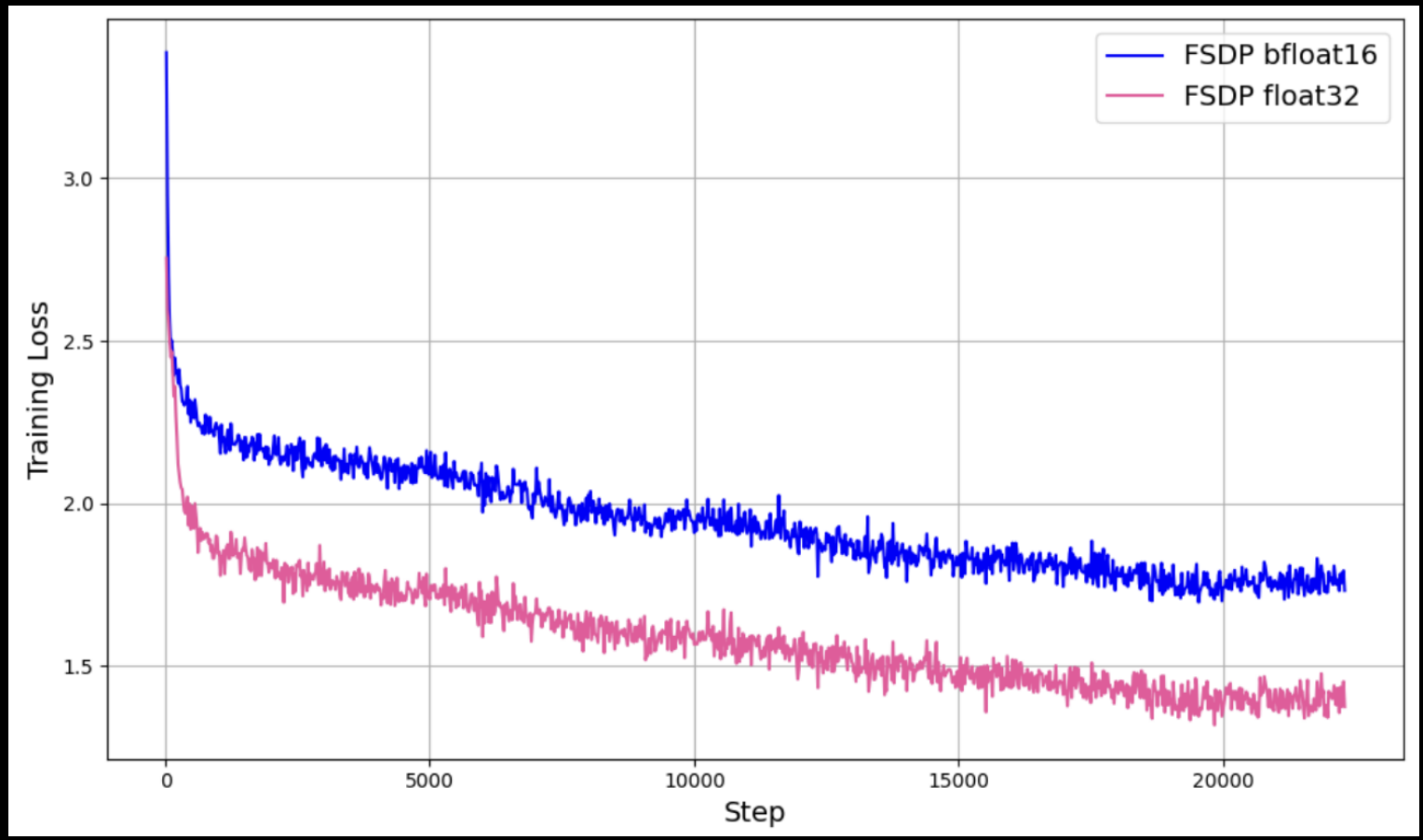
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Quantization at Training

How much precision is needed for training?

e.g., recall batch normalization causes values to center on 0 and range roughly between -1 and 1

What are risks of using fewer bits?



[Dietke, CVPR25]

Quantization at Training

What are risks and benefits of using fewer bits (e.g., when done with pruning)?

How much precision is needed for training?

e.g., recall batch normalization causes values to center on 0 and range roughly between -1 and 1

	Actual No. Of Parameters/ size	Actual Top-5 Error Rate (%)	Method Type	Para./size after compression	Compression Achieved	Top-5 Error Rate (%) after Compression	Speedup Achieved
AlexNet	61M/240 MB	19.7	Pruning	6.7M	9x	19.67	3x
			Pruning and quantization	6.9 MB	35x	19.7	3x
VGG16	138M/512 MB	10.4	Pruning	10.3M	13x	10.88	5x
			Pruning and quantization	11.3 MB	49x	10.91	3x to 4x

https://xailient.com/blog/4-popular-model-compression-techniques-explained/

Quantization Options

- e.g.,

	A100 80GB PCIe	A100 80GB SXM
FP64 (Exceeds default)	9.7 TFLOPS	
FP64 Tensor Core (Exceeds default)	19.5 TFLOPS	
FP32 (default)	19.5 TFLOPS	
Tensor Float 32 (TF32)	156 TFLOPS	312 TFLOPS*
BFLOAT16 Tensor Core	312 TFLOPS	624 TFLOPS*
FP16 Tensor Core	312 TFLOPS	624 TFLOPS*
INT8 Tensor Core	624 TOPS	1248 TOPS*

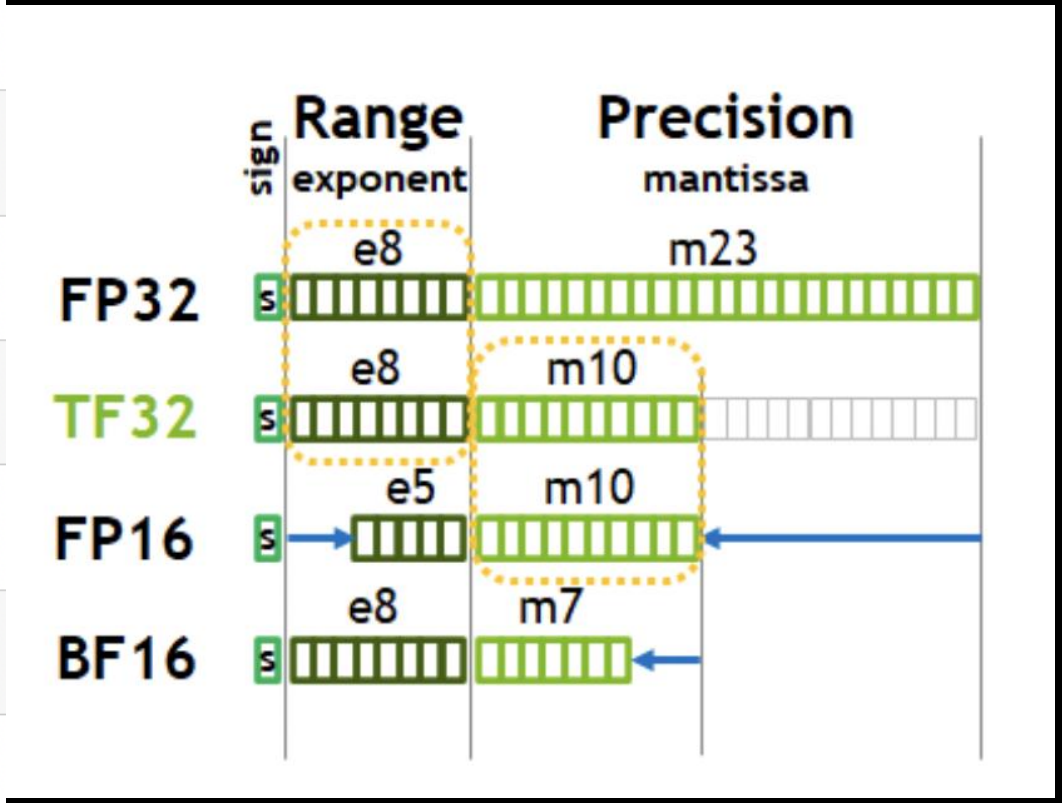
How much is the speed-up?
(without *sparsity)

https://www.nvidia.com/en-us/data-center/a100/

Quantization Options

- e.g.,

FP64
FP64 Tensor Core
FP32
Tensor Float 32 (TF32)
BFLOAT16 Tensor Core
FP16 Tensor Core
INT8 Tensor Core



Greater speed-ups from greater precision cuts!

https://images.nvidia.com/aem-dam/en-zz/Solutions/data-center/nvidia-ampere-architecture-whitepaper.pdf

Quantization Options: A100 vs H200

NVIDIA H200 Tensor Core GPU		A100 80GB PCIe
Form Factor	**H200 SXM[1]**	
FP64	34 TFLOPS	9.7 TFLOPS
FP64 Tensor Core	67 TFLOPS	19.5 TFLOPS
FP32	67 TFLOPS	19.5 TFLOPS
TF32 Tensor Core	989 TFLOPS[2]	156 TFLOPS | 312 TFLOPS*
BFLOAT16 Tensor Core	1,979 TFLOPS[2]	312 TFLOPS | 624 TFLOPS*
FP16 Tensor Core	1,979 TFLOPS[2]	312 TFLOPS | 624 TFLOPS*
INT8 Tensor Core	3,958 TFLOPS[2]	624 TOPS | 1248 TOPS*

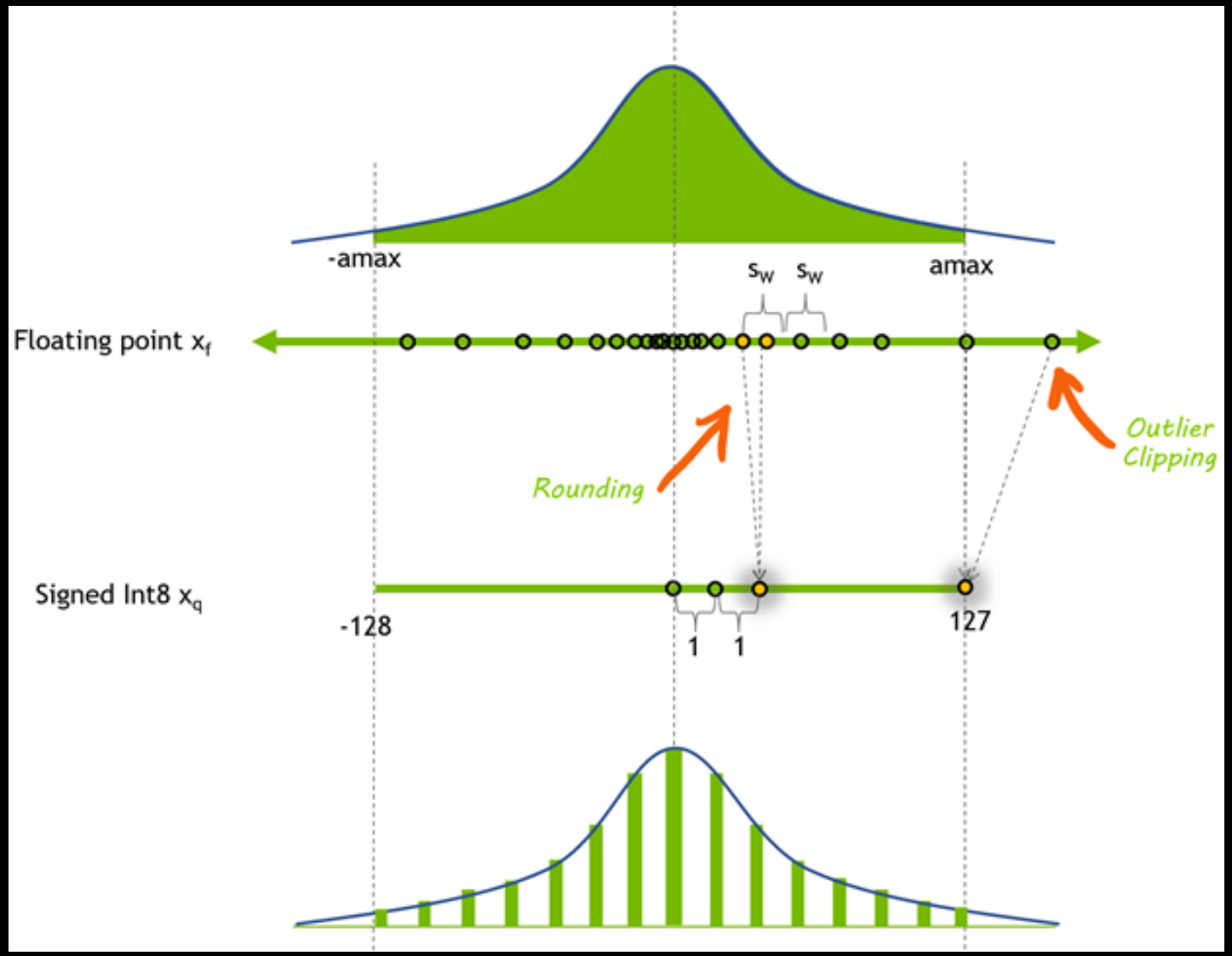
Use for training?

https://www.fibermall.com/blog/why-gpu-require-hbm.htm; https://www.nvidia.com/en-us/data-center/a100/

Quantization at Inference

e.g., scale to [-128, 127] with 8 bits

- Outliers set to minimum and maximum

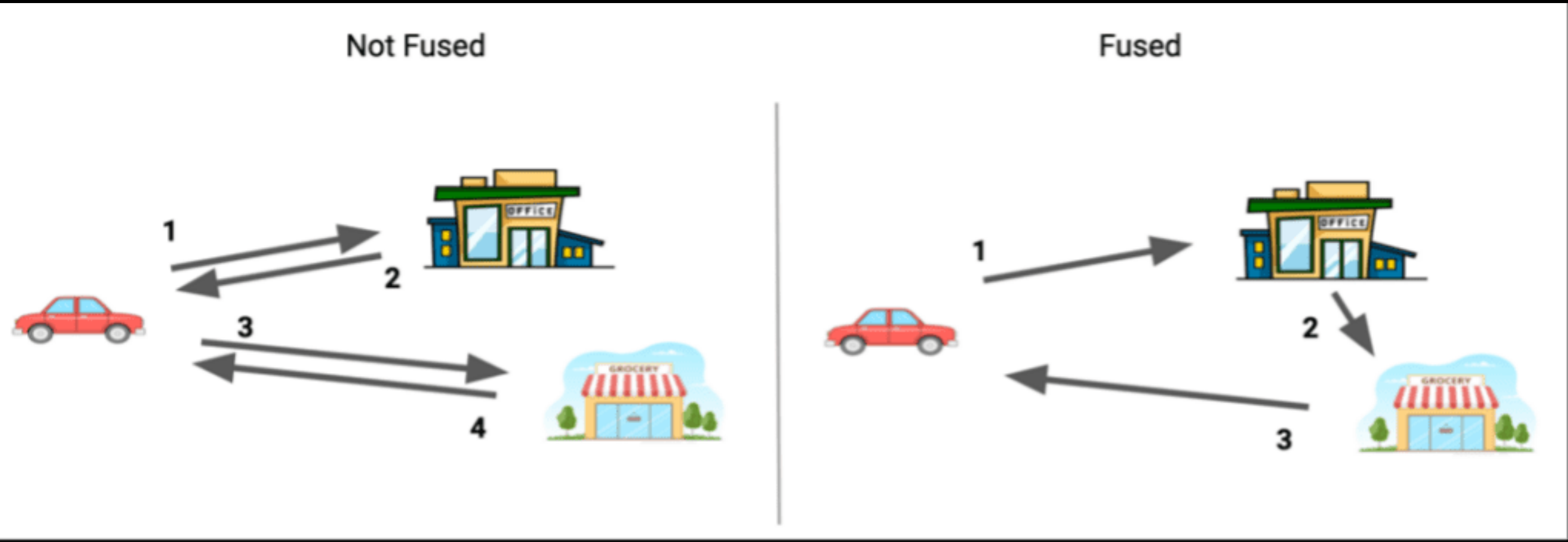


https://developer.nvidia.com/blog/achieving-fp32-accuracy-for-int8-inference-using-quantization-aware-training-with-tensorrt/

Platform Features Enabling Faster, More Efficient GPU Training

- Quantization: reduce precision
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Idea: Fuse Tasks to Reduce Overhead



(e.g., go to school, return home, buy groceries)

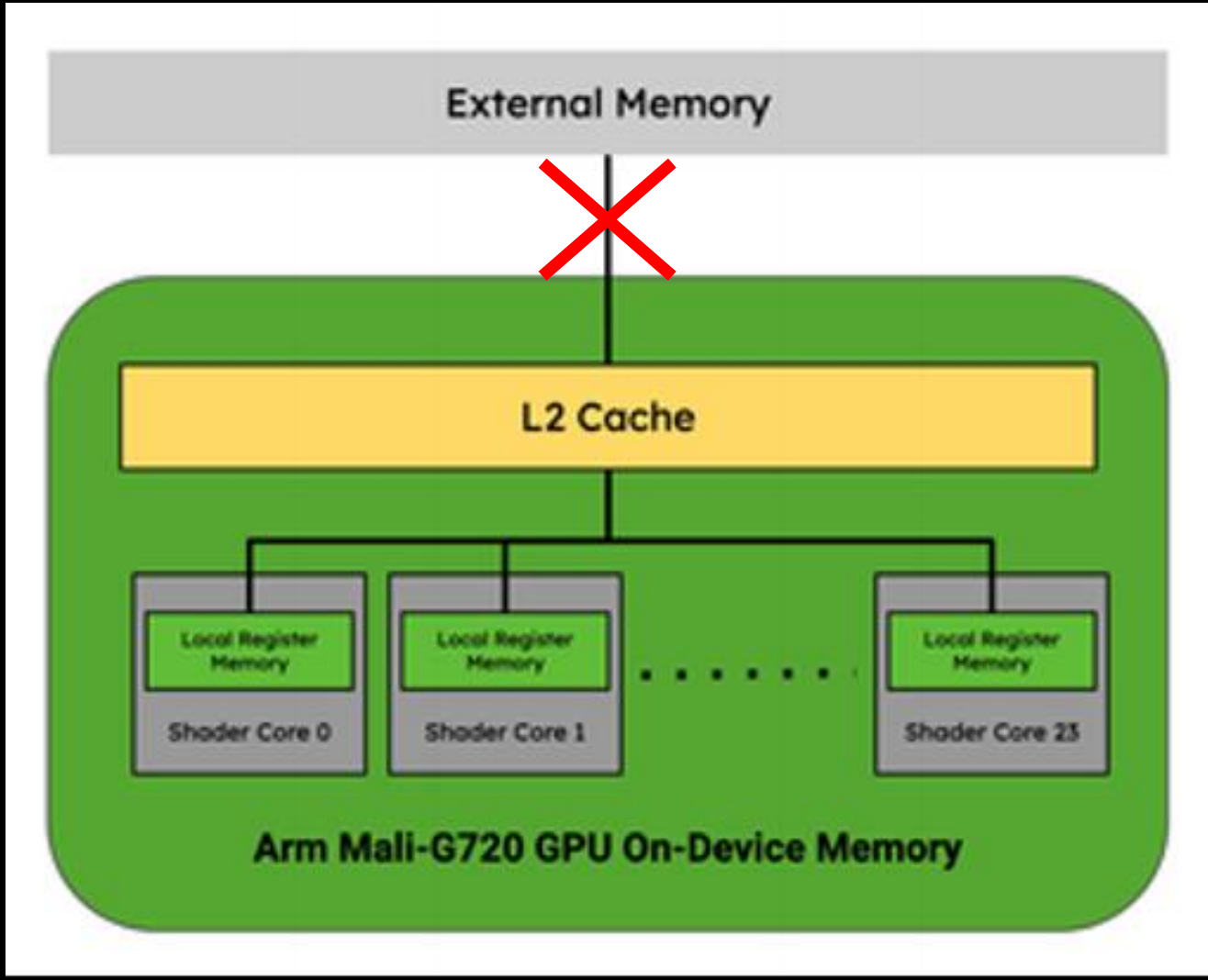
(e.g., go to school and then buy groceries)

https://quadric.io/2023/09/13/how-to-unlock-the-power-of-operator-fusion-to-accelerate-ai/

Approach: Fuse Tasks to Reduce Overhead

Avoid shuttling data back-and-forth to external memory by determining what operations to fuse so relevant information stays in limited memory close to the GPU

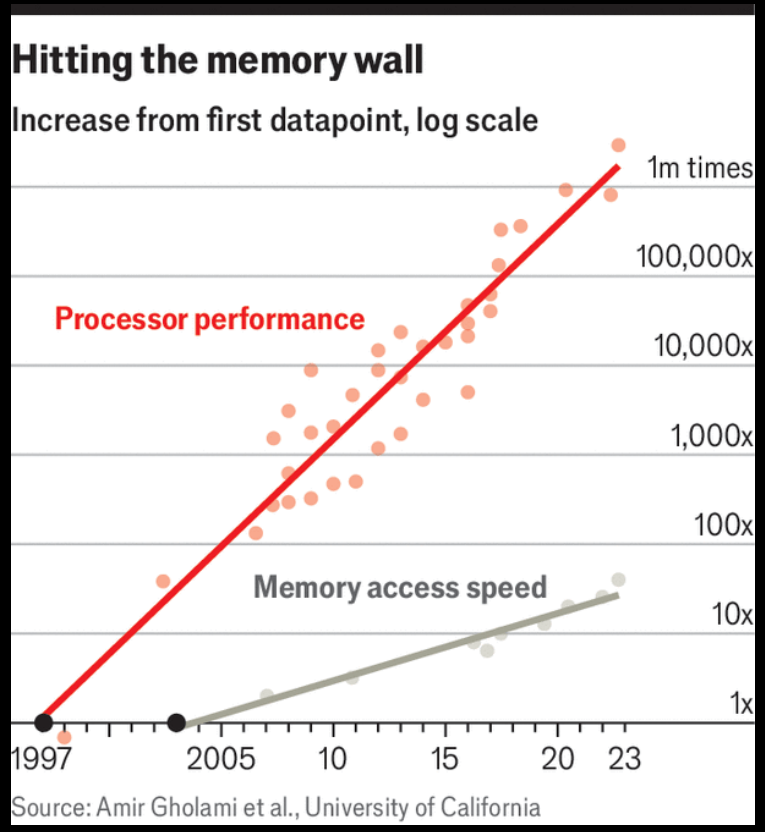
- When would you prefer L2 cache versus Load Register Memory?



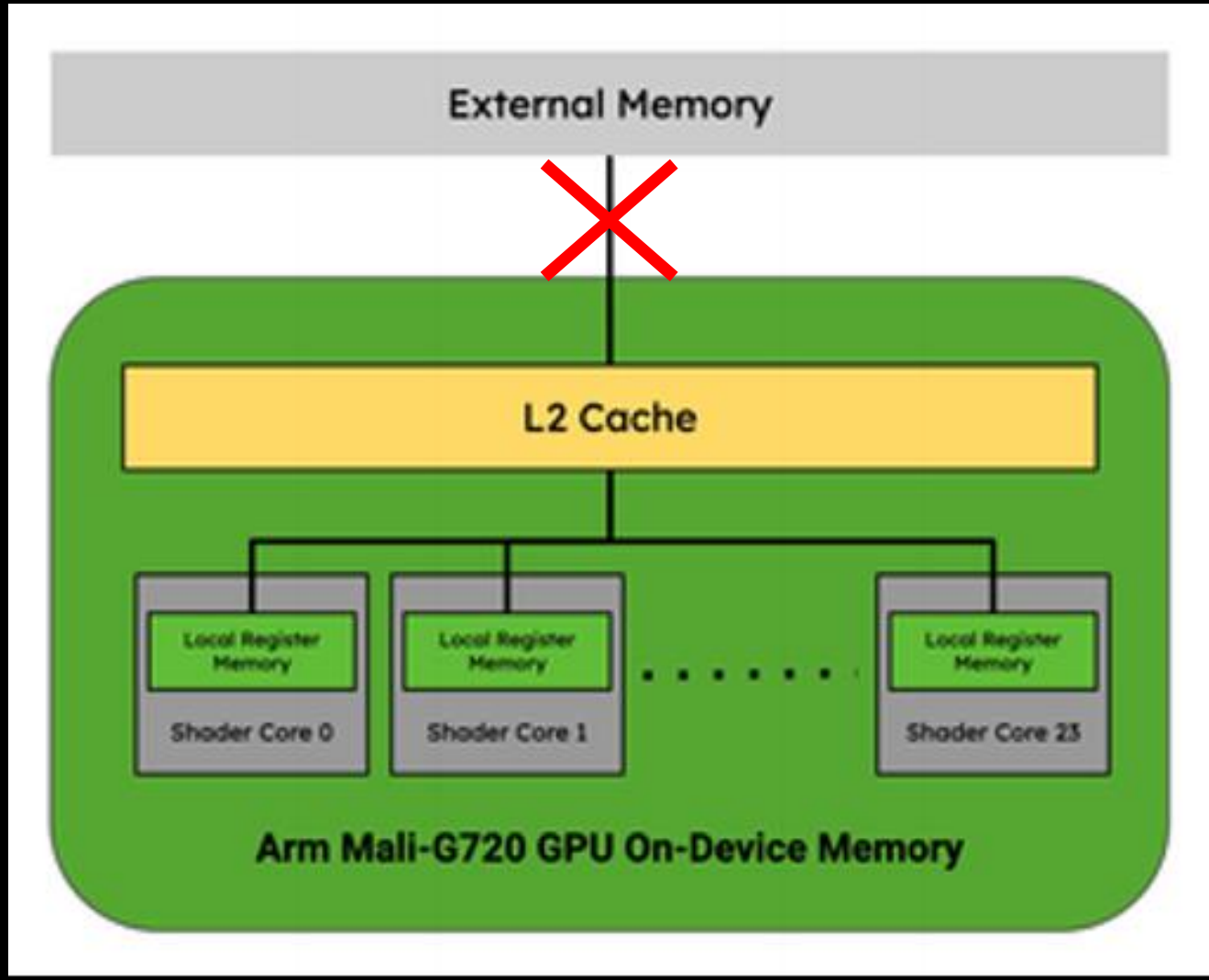
https://quadric.io/2023/09/13/how-to-unlock-the-power-of-operator-fusion-to-accelerate-ai/

Approach: Fuse Tasks to Reduce Overhead

Especially important given HW trends: processor performance (tripling every two years) exceeds that of memory access speed (~half as much)



(Economist, Sep 21, 2024)

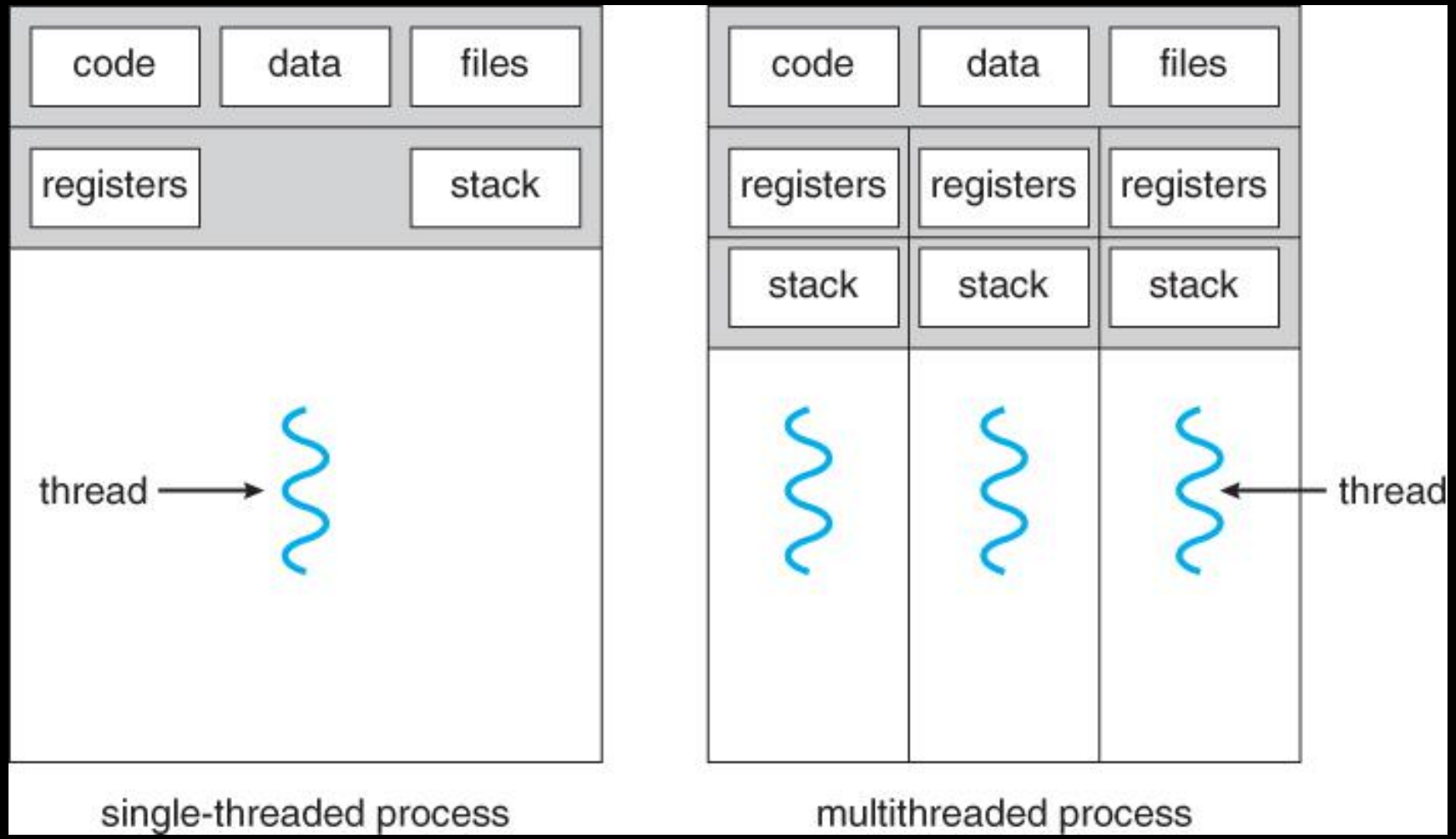


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Recall Multi-Thread Programming



https://www.cs.uic.edu/~jbell/CourseNotes/OperatingSystems/4_Threads.html

Distribute Neural Network Computations Across Multiple Devices (e.g., GPUs, CPUs)

- Key questions:
  - What tasks to delegate to each device?
  - What communication to facilitate across devices and how to do this?
- Popular Methods:
  - Distributed Data Parallel (DDP): each “machine” holds copy of model to perform forward and backward passes on different data subsets, with cross-machine communication (i.e., check-pointing) to identify average gradients across machines
    - If you had 4 GPUs, approximately what amount of speed-up might you expect?
  - Distributed Lower-Communication Training (DiLoCo): distributes training to multiple “islands” of a massive cluster (e.g., 100,000 GPUs), with regular checkpointing in each and less checkpointing across them, enabling devices to come from different sources (e.g., locations, companies, countries)
  - Zero Redundancy Optimizer (ZeRO): supports parallelism *without* code changes; partitions model training states (e.g., weights, gradients) across devices, enabling training of very large models (e.g., trillions of parameters)

Summary: Platform Features Enabling Faster, More Efficient GPU Training

- Quantization: reduce precision
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Today’s Topics

- The quest for more data and faster machines
- Paradigm shift: learned representations thrive with large labelled datasets
- Lucky break: gradient descent meets gaming hardware
- Tipping point: open-source platforms catalyze widespread engagement
- Programming tutorial

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The End